

245 Industrial Parkway North  
Aurora, ON L4G 4C4  
Phone: (905) 669-7999  
Fax: 905-669-0268

**Date: September 29, 2026 Job No.: 2025-048**

To: Diamond Schmitt

Attn: Ashish Kumar

C.C.: Project File, A. Stephens, M. Sinno, K. Schaeffer, M. Diab, A. Haidar, H. Singh, D. Patel, A. Hernandez,

Re: **Submittal 23 83 00-2-R1-Revised Radiant Slab Heating and Cooling System Shop Drawings**

**WE ARE SENDING YOU** 1 electronic copy via email of the following items:

| ZERO CARBON DOCUMENTS | SHOP DRAWINGS | PRODUCT INFORMATION | SAMPLES / MOCK UP | SUBSTITUTION REQUEST | WARRANTY |  |  |
|-----------------------|---------------|---------------------|-------------------|----------------------|----------|--|--|
|                       | X             | X                   |                   |                      |          |  |  |

| NO. | DATE        | REV # | DESCRIPTION                                                                   |
|-----|-------------|-------|-------------------------------------------------------------------------------|
| 1   | Sept. 22/26 | 0     | MSB Transmittal                                                               |
| 2   | Sept. 22/26 | 0     | Dwgs 1 through 3: Interior Radiant Flooring Heating Layout Plans and Schedule |
| 3   | Sept. 22/26 | 0     | Dwg. 1: Exterior Snow Melting System                                          |
| 4   | Sept. 22/26 | 0     | Wirsbo hePEX Coils PDS                                                        |
| 5   | Sept. 22/26 | 0     | Stainless-Steel Manifold Assembly, 1 1/4" With Flow Meter PDS                 |
| 6   | Sept. 22/26 | 0     | Stainless-Steel Manifold Assembly, 1" With Flow Meter PDS                     |
| 7   | Sept. 22/26 | 0     | Manifold Wall Cabinets, Recessed Mount PDS                                    |
| 8   | Sept. 22/26 | 0     | Valve Manifold Cabinet – Surface Mount PDS                                    |
| 9   | Sept. 22/26 | 0     | ProPEX Manifold Fitting Assemblies PDS                                        |
| 10  | Sept. 22/26 | 0     | PVC Bend Supports PDS                                                         |
| 11  | Sept. 22/26 | 0     | Snow Melt Control PDS                                                         |
| 12  | Sept. 22/26 | 0     | ESBE Type F 3 & 4-Way Rotary Valves PDS                                       |

|    |             |   |                                              |
|----|-------------|---|----------------------------------------------|
| 14 | Sept. 22/26 | 0 | ESBE Series 90 Electronic Motor Actuator PDS |
|----|-------------|---|----------------------------------------------|

THESE ARE TRANSMITTED as checked below:

☒ For approval

☐ For signature

☐ For your information

☐ As requested

☐ For review and comment

REMARKS:

---

*Hari Singh*

**Signature**

This transmission, including any accompanying attachments, contains information that may be privileged and/or exempt from disclosure under applicable law, and is intended only for the use of the recipient(s) named above. Any distribution, review, dissemination, or copying of the contents of this communication by anyone other the intended recipient(s) is strictly prohibited. If you receive this communication in error, please notify the sender immediately and destroy the copy you have received. Thank you.



Kolostat Inc.  
1200 Stellar Drive  
Newmarket, Ontario L3Y 7B9  
P: 905-898-7979

Project: 25-013 Docksteader PRPS (Peel Region  
Paramedic Services)  
6 Docksteader Road  
Brampton, Ontario L6R 0A9

Submittal #66.2 - Radiant Floor Heating and Snow Melt System  
23 83 00 - RADIANT HEATING AND COOLING SYSTEMS

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------|
| Revision               | 2                                                                                                                                                                                                                                                                                                                                                                                                                                    | Submittal Manager | Rajwant Kaur Khattrra (Kolostat Inc - Newmarket) |
| Status                 | Submitted                                                                                                                                                                                                                                                                                                                                                                                                                            | Date Created      | Sep 16, 2026                                     |
| Issue Date             | Sep 16, 2026                                                                                                                                                                                                                                                                                                                                                                                                                         | Spec Section      | 23 83 00 - RADIANT HEATING AND COOLING SYSTEMS   |
| Responsible Contractor | The Morgan Group                                                                                                                                                                                                                                                                                                                                                                                                                     | Received From     | Craig Stoneman (The Morgan Group)                |
| Received Date          |                                                                                                                                                                                                                                                                                                                                                                                                                                      | Submit By         |                                                  |
| Final Due Date         | Sep 28, 2026                                                                                                                                                                                                                                                                                                                                                                                                                         | Lead Time         |                                                  |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                      | Cost Code         |                                                  |
| Location               |                                                                                                                                                                                                                                                                                                                                                                                                                                      | Type              | Shop drawing                                     |
| Submittal Package      |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                  |
| Approvers              | Alvin Hernandez (Matheson Constructors)                                                                                                                                                                                                                                                                                                                                                                                              |                   |                                                  |
| Ball in Court          | Alvin Hernandez (Matheson Constructors)                                                                                                                                                                                                                                                                                                                                                                                              |                   |                                                  |
| Distribution           | Ali Haider (Matheson Constructors), Alvin Hernandez (Matheson Constructors), Brad Lilleberg (Kolostat Inc - Newmarket), Craig Stoneman (The Morgan Group), DPRS Email (Kolostat Inc - Newmarket), Ghislain Boudreau (Kolostat Inc - Quebec), Hari Singh (Matheson Constructors), Kathy Schaeffer (Matheson Constructors), Manny Oliveira (Kolostat Inc - Newmarket), Marwan Diab (Matheson Constructors), Steve Pigram (Next Supply) |                   |                                                  |
| Description            | REASON FOR REVISION-<br>ADDED MANIFOLD CABINET DETAILS                                                                                                                                                                                                                                                                                                                                                                               |                   |                                                  |
| LEAD TIME              |                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                                                  |

Submittal Workflow

| Name                            | Sent Date    | Due Date     | Returned Date | Response | Attachments                                                      |
|---------------------------------|--------------|--------------|---------------|----------|------------------------------------------------------------------|
| General Information Attachments |              |              |               |          | <a href="#">SD 66 R2 - Radiant Floor And Snowmelt System.pdf</a> |
| Alvin Hernandez                 | Sep 16, 2026 | Sep 28, 2026 |               | Pending  |                                                                  |

Consultant: This submittal is to be reviewed by the Architect and Mechanical Engineer.

MATHESON CONSTRUCTORS

Reviewed

Reviewed as Noted

Revise and Resubmit

Not Reviewed

Date: 2026-09-22

Reviewed by:

This review by MATHESON CONSTRUCTORS LIMITED is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of engineering elements which form part of the contract documents prepared by others. This review shall not mean that MATHESON CONSTRUCTORS LIMITED approves the detailed design inherent in the shop drawings, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the shop drawings or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and coordinated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.

Project Name  
Dockstader PRS

APPROVED

REJECTED

REVIEWED

REVIS & RESUBMIT

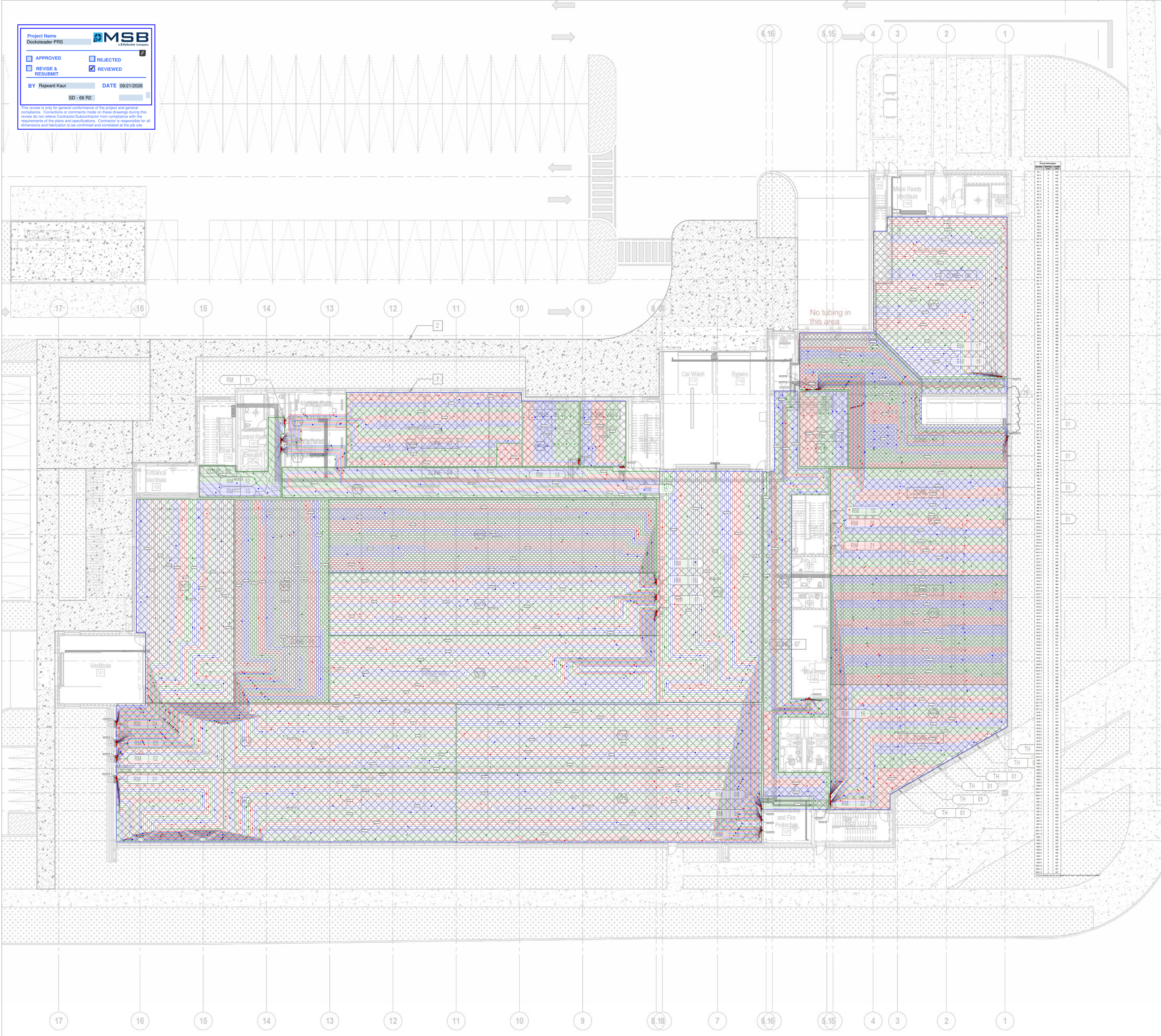
REVIEWED

BY Rajwant Kaur

DATE 09/21/2026

SD - 66 R2

This review is only for general conformance of the project and general compliance. Corrections or comments made on these drawings during this review do not relieve Contractor/Subcontractor from compliance with the requirements of the plans and specifications. Contractor is responsible for all dimensions and fabrication to be confirmed and correlated at the job site.



Construction Services

DISCLAIMER:  
Note: Details of the tubing bend radius have been simplified for clarity. Consult tubing manufacturer for specific bend radius recommendations.

PROJECT:  
**Dockstader PRPS**  
CUSTOMER:

PROJECT NO.:  
**54906H R2**

SCALE:  
**3/32"=1'**

DRAWING NAME:  
**Level 1**

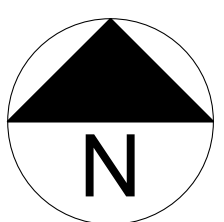
DRAWN BY:  
**Leah Hayden**

Created Using LoopCAD 2026 26.0.0095 (9/18/2026)

DATE:  
**9/18/2026**

REVISIONS

| No | Desc                                                   | Date      |
|----|--------------------------------------------------------|-----------|
| R1 | Relocate & reroute for manifolds 12,14,16,19,20,21,22. | 8/25/2026 |
| R2 | Relocate & reroute for manifolds 23,25,27,28,29,30     | 9/18/2026 |



DISCLAIMER:  
Note: Details of the tubing bend radius have been simplified for clarity. Consult tubing manufacturer for specific bend radius recommendations.

PROJECT:  
**Dockstader PRPS**  
CUSTOMER:

PROJECT NO.:  
**54906H R2**

SCALE:  
**1/8"=1'**

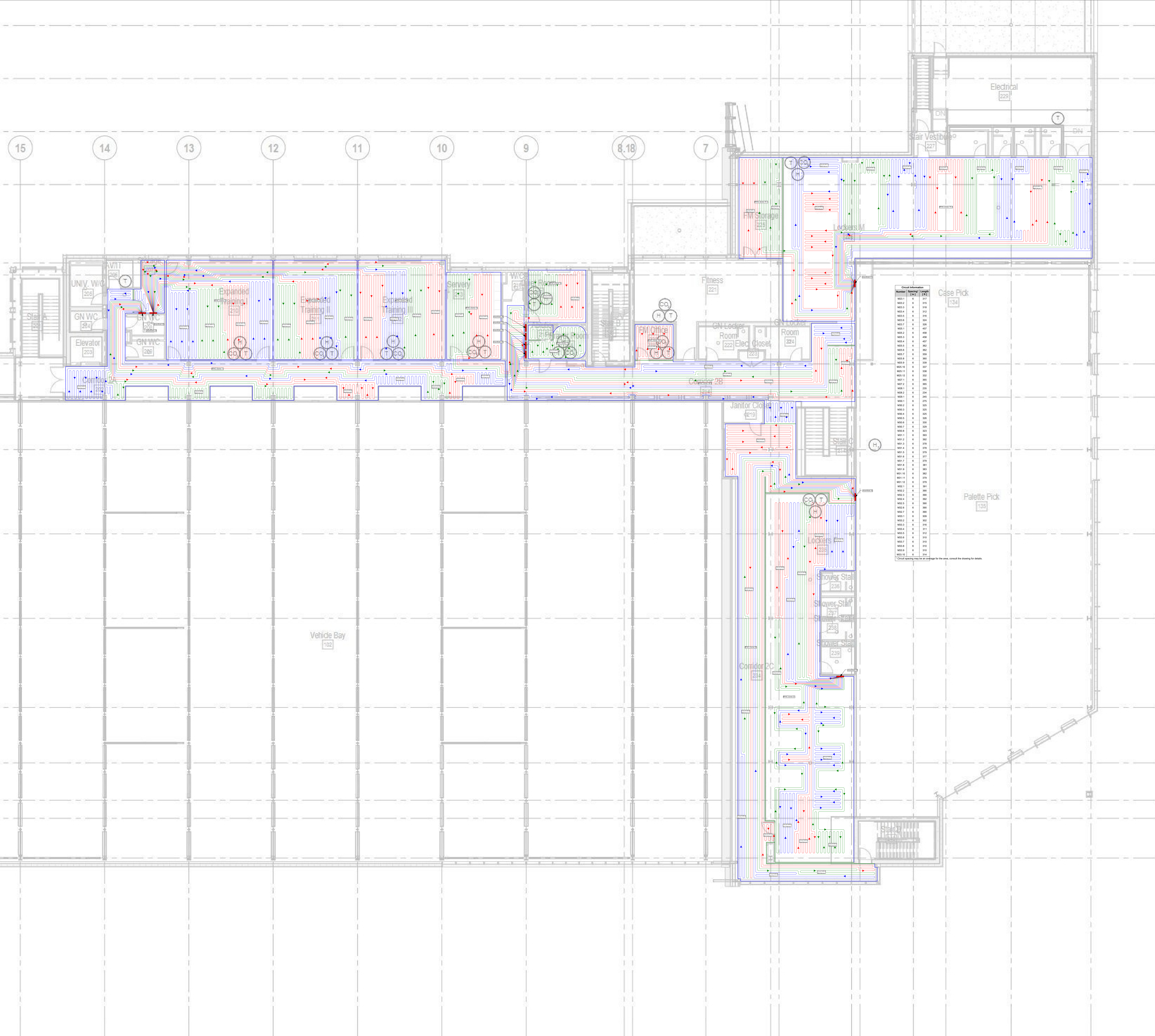
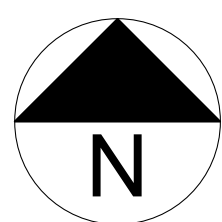
DRAWING NAME:  
**Level 2**

DRAWN BY:  
**Leah Hayden**  
Created Using LoopCAD 2026 26.0.0095 (9/18/2026)

DATE:  
**9/18/2026**

REVISIONS

| No | Desc                                                   | Date      |
|----|--------------------------------------------------------|-----------|
| R1 | Relocate & reroute for manifolds 12,14,18,19,20,21,22. | 8/25/2026 |
| R2 | Relocate & reroute for manifolds 23,25,27,28,29,30     | 9/18/2026 |



DISCLAIMER:  
Note: Details of the tubing bend radius have been simplified for clarity. Consult tubing manufacturer for specific bend radius recommendations.

PROJECT: **Dockstader PRPS**

CUSTOMER:

PROJECT NO.: **54906H R2**

SCALE:  
**Not To Scale**

DRAWING NAME:  
**Schedule**

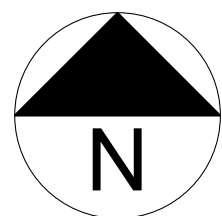
DRAWN BY:  
**Leah Hayden**

Created Using LoopCAD 2026 26.0.0095 (9/18/2026)

DATE:  
**9/18/2026**

REVISIONS

| No | Desc                                                   | Date      |
|----|--------------------------------------------------------|-----------|
| R1 | Relocate & reroute for manifolds 12,14,16,19,20,21,22. | 8/25/2026 |
| R2 | Relocate & reroute for manifolds 23,25,27,28,29,30     | 9/18/2026 |



Project Name: Dockstader PRPS  
Project No. 54906-R2

| Radiant Schedule |                   |                  |              |              |                 |          |              |              |                |                |                       |                 |                   |             |                |            |         |         |         |         |            |            |              |               |              |              |                |                |                       |                 |                   |             |                |            |         |         |         |         |            |       |            |              |               |      |              |          | Notes       |            |     |
|------------------|-------------------|------------------|--------------|--------------|-----------------|----------|--------------|--------------|----------------|----------------|-----------------------|-----------------|-------------------|-------------|----------------|------------|---------|---------|---------|---------|------------|------------|--------------|---------------|--------------|--------------|----------------|----------------|-----------------------|-----------------|-------------------|-------------|----------------|------------|---------|---------|---------|---------|------------|-------|------------|--------------|---------------|------|--------------|----------|-------------|------------|-----|
| Zone Name        | Active Area (ft²) | Active Area (m²) | Manifold Tag | Manufacturer | Manifold Type   | Glycol % | Cooling      |              |                |                |                       |                 |                   |             |                |            |         |         |         |         | Heating    |            |              |               |              |              |                |                |                       |                 |                   |             |                |            | Tubing  |         |         |         |            | Notes |            |              |               |      |              |          |             |            |     |
|                  |                   |                  |              |              |                 |          | Setpoint (F) | Setpoint (C) | Fl Surface (F) | Fl Surface (C) | Capacity (Btu/hr/ft²) | Capacity (W/m²) | Gain (Btu/hr/ft²) | Gain (W/m²) | Total (Btu/hr) | Total (kw) | EWT (F) | EWT (C) | LWT (F) | LWT (C) | Flow (GPM) | Flow (L/s) | HD Loss (FT) | HD Loss (Kpa) | Setpoint (F) | Setpoint (C) | Fl Surface (F) | Fl Surface (C) | Capacity (Btu/hr/ft²) | Capacity (W/m²) | Loss (Btu/hr/ft²) | Loss (W/m²) | Total (Btu/hr) | Total (kw) | EWT (F) | EWT (C) | LWT (F) | LWT (C) | Flow (GPM) |       | Flow (L/s) | HD Loss (FT) | HD Loss (Kpa) | Type | Spacing (in) | No Loops | Length (ft) | Length (m) |     |
| RFHC-Zone 1-1    | 3016              | 280              | RFM-1        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 29255          | 8.6        | 59      | 15.0    | 65      | 18.3    | 9.8        | 0.6        | 8.7          | 25.8          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 97417          | 28.5       | 110     | 43.3    | 90      | 32.2    | 9.7        | 0.6   | 8.0        | 23.9         | 5/8" hePEX    | 9"   | 228          | 12       | 443         | 135        | 1.5 |
| RFHC-Zone 1-2    | 2121              | 197              | RFM-2        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 20574          | 6.0        | 59      | 15.0    | 65      | 18.3    | 6.9        | 0.4        | 4.0          | 12.0          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 88508          | 20.1       | 110     | 43.3    | 90      | 32.2    | 6.9        | 0.4   | 3.7        | 11.1         | 5/8" hePEX    | 9"   | 228          | 12       | 363         | 111        | 1.5 |
| RFHC-Zone 1-3    | 3199              | 297              | RFM-3        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 31030          | 9.1        | 59      | 15.0    | 65      | 18.3    | 10.3       | 0.7        | 9.6          | 28.5          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 103328         | 30.3       | 110     | 43.3    | 90      | 32.2    | 10.3       | 0.7   | 8.8        | 28.4         | 5/8" hePEX    | 9"   | 228          | 12       | 439         | 134        | 1.5 |
| RFHC-Zone 1-4    | 2711              | 252              | RFM-4        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 26297          | 7.7        | 59      | 15.0    | 65      | 18.3    | 8.8        | 0.6        | 8.1          | 24.1          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 87565          | 25.7       | 110     | 43.3    | 90      | 32.2    | 8.8        | 0.6   | 7.5        | 22.3         | 5/8" hePEX    | 9"   | 228          | 12       | 380         | 116        | 1.5 |
| RFHC-Zone 1-5    | 3255              | 302              | RFM-5        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 31574          | 9.3        | 59      | 15.0    | 65      | 18.3    | 10.5       | 0.7        | 9.2          | 27.6          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 105137         | 30.8       | 110     | 43.3    | 90      | 32.2    | 10.5       | 0.7   | 8.6        | 25.6         | 5/8" hePEX    | 9"   | 228          | 12       | 396         | 121        | 1.5 |
| RFHC-Zone 1-6    | 2372              | 220              | RFM-6        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 23080          | 6.7        | 59      | 15.0    | 65      | 18.3    | 7.7        | 0.5        | 4.5          | 13.4          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 78616          | 22.4       | 110     | 43.3    | 90      | 32.2    | 7.7        | 0.5   | 4.2        | 12.5         | 5/8" hePEX    | 9"   | 228          | 12       | 314         | 96         | 1.5 |
| RFHC-Zone 1-7    | 2786              | 258              | RFM-7        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 28809          | 8.4        | 59      | 15.0    | 65      | 18.3    | 9.6        | 0.6        | 8.1          | 24.1          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 95931          | 28.1       | 110     | 43.3    | 90      | 32.2    | 9.6        | 0.6   | 7.5        | 22.3         | 5/8" hePEX    | 9"   | 228          | 12       | 416         | 127        | 1.5 |
| RFHC-Zone 1-8    | 3700              | 344              | RFM-8        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 35890          | 10.5       | 59      | 15.0    | 65      | 18.3    | 12.0       | 0.8        | 12.6         | 37.6          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 119510         | 35.0       | 110     | 43.3    | 90      | 32.2    | 12.0       | 0.8   | 11.7       | 34.9         | 5/8" hePEX    | 9"   | 228          | 12       | 447         | 136        | 1.5 |
| RFHC-Zone 1-9    | 2736              | 254              | RFM-9        | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 26539          | 7.8        | 59      | 15.0    | 65      | 18.3    | 8.8        | 0.6        | 6.5          | 19.5          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 86373          | 25.9       | 110     | 43.3    | 90      | 32.2    | 8.8        | 0.6   | 6.1        | 18.1         | 5/8" hePEX    | 9"   | 228          | 12       | 377         | 115        | 1.5 |
| RFHC-Zone 1-10   | 2201              | 204              | RFM-10       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 68             | 19.7           | 9.2                   | 29.0            | -0.5              | -1.6        | 21350          | 6.3        | 59      | 15.0    | 65      | 18.3    | 7.1        | 0.4        | 4.1          | 12.2          | 65           | 18.3         | 81             | 27.2           | 31.0                  | 97.8            | -1.3              | -4.1        | 71092          | 20.8       | 110     | 43.3    | 90      | 32.2    | 7.1        | 0.4   | 3.8        | 11.3         | 5/8" hePEX    | 9"   | 228          | 12       | 336         | 102        | 1.5 |
| RFHC-Zone 2      | 374               | 35               | RFM-11       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.5           | 9.8                   | 30.9            | -0.5              | -1.6        | 3652           | 1.1        | 59      | 15.0    | 65      | 18.3    | 1.3        | 0.1        | 3.8          | 11.4          | 70           | 21.1         | 85             | 29.4           | 29.0                  | 91.5            | -1.5              | -4.7        | 11407          | 3.3        | 110     | 43.3    | 90      | 32.2    | 1.1        | 0.1   | 2.8        | 8.4          | 5/8" hePEX    | 6"   | 152          | 2        | 375         | 114        | 2.6 |
| RFHC-Zone 3      | 1604              | 149              | RFM-12       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 71             | 21.4           | 5.4                   | 17.0            | -0.5              | -1.6        | 9464           | 2.8        | 59      | 15.0    | 65      | 18.3    | 3.2        | 0.2        | 1.4          | 4.0           | 70           | 21.1         | 77             | 25.1           | 14.1                  | 44.5            | -1.6              | -5.0        | 25183          | 7.4        | 110     | 43.3    | 90      | 32.2    | 2.5        | 0.2   | 0.8        | 2.5          | 5/8" hePEX    | 6"   | 152          | 10       | 390         | 119        | 2.7 |
| RFHC-Zone 4      | 1405              | 131              | RFM-13       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.5           | 9.8                   | 30.9            | -0.5              | -1.6        | 14472          | 4.2        | 59      | 15.0    | 65      | 18.3    | 4.8        | 0.3        | 4.1          | 12.2          | 70           | 21.1         | 85             | 29.4           | 29.0                  | 91.5            | -1.5              | -4.7        | 42853          | 12.6       | 110     | 43.3    | 90      | 32.2    | 4.3        | 0.3   | 3.0        | 9.0          | 5/8" hePEX    | 6"   | 152          | 8        | 412         | 126        | 2.6 |
| RFHC-Zone 5      | 542               | 50               | RFM-14       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 71             | 21.4           | 5.4                   | 17.0            | -0.5              | -1.6        | 3198           | 0.9        | 59      | 15.0    | 65      | 18.3    | 1.1        | 0.1        | 1.4          | 4.0           | 70           | 21.1         | 77             | 25.1           | 14.1                  | 44.5            | -1.6              | -5.0        | 8509           | 2.5        | 110     | 43.3    | 90      | 32.2    | 0.9        | 0.1   | 0.8        | 2.4          | 5/8" hePEX    | 6"   | 152          | 4        | 369         | 112        | 2.7 |
| RFHC-Zone 6      | 339               | 31               | RFM-15       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.5           | 9.8                   | 30.9            | -0.5              | -1.6        | 3492           | 1.0        | 59      | 15.0    | 65      | 18.3    | 1.2        | 0.1        | 1.1          | 3.2           | 70           | 21.1         | 85             | 29.4           | 29.0                  | 91.5            | -1.5              | -4.7        | 10340          | 3.0        | 110     | 43.3    | 90      | 32.2    | 1.0        | 0.1   | 0.8        | 2.4          | 5/8" hePEX    | 6"   | 152          | 3        | 236         | 72         | 2.6 |
| RFHC-Zone 7      | 1699              | 158              | RFM-16       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.5           | 9.8                   | 30.9            | -0.5              | -1.6        | 17500          | 5.1        | 59      | 15.0    | 65      | 18.3    | 5.8        | 0.4        | 3.2          | 9.6           | 70           | 21.1         | 85             | 29.4           | 29.0                  | 91.5            | -1.5              | -4.7        | 51620          | 15.2       | 110     | 43.3    | 90      | 32.2    | 5.2        | 0.3   | 2.4        | 7.1          | 5/8" hePEX    | 6"   | 152          | 11       | 341         | 104        | 1.6 |
| RFHC-Zone 8      | 2520              | 234              | RFM-17       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.7           | 9.3                   | 29.3            | -0.5              | -1.6        | 24896          | 7.2        | 59      | 15.0    | 65      | 18.3    | 8.2        | 0.5        | 5.1          | 15.2          | 70           | 21.1         | 84             | 28.8           | 27.0                  | 85.2            | -1.3              | -4.1        | 71316          | 20.9       | 110     | 43.3    | 90      | 32.2    | 7.1        | 0.4   | 3.6        | 10.9         | 5/8" hePEX    | 9"   | 228          | 12       | 310         | 94         | 1.8 |
| RFHC-Zone 9      | 463               | 43               | RFM-18       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.5           | 9.8                   | 30.9            | -0.5              | -1.6        | 4769           | 1.4        | 59      | 15.0    | 65      | 18.3    | 1.6        | 0.1        | 2.4          | 7.1           | 70           | 21.1         | 85             | 29.4           | 29.0                  | 91.5            | -1.5              | -4.7        | 14122          | 4.1        | 110     | 43.3    | 90      | 32.2    | 1.4        | 0.1   | 1.7        | 5.2          | 5/8" hePEX    | 6"   | 152          | 3        | 317         | 97         | 1.6 |
| RFHC-Zone 10-1   | 1502              | 140              | RFM-19       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.2           | 10.4                  | 32.8            | -0.5              | -1.6        | 16372          | 4.8        | 59      | 15.0    | 65      | 18.3    | 5.5        | 0.3        | 2.3          | 7.0           | 70           | 21.1         | 86             | 30.1           | 31.4                  | 99.1            | -1.5              | -4.7        | 48416          | 14.5       | 110     | 43.3    | 90      | 32.2    | 4.9        | 0.3   | 1.8        | 5.4          | 5/8" hePEX    | 6"   | 152          | 12       | 297         | 91         | 1.5 |
| RFHC-Zone 10-2   | 2982              | 277              | RFM-20       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.2           | 10.4                  | 32.8            | -0.5              | -1.6        | 32504          | 9.5        | 59      | 15.0    | 65      | 18.3    | 10.8       | 0.7        | 11.8         | 35.4          | 70           | 21.1         | 86             | 30.1           | 31.4                  | 99.1            | -1.5              | -4.7        | 98108          | 28.7       | 110     | 43.3    | 90      | 32.2    | 9.8        | 0.6   | 9.1        | 27.3         | 5/8" hePEX    | 6"   | 152          | 12       | 532         | 162        | 1.5 |
| RFHC-Zone 10-3   | 2946              | 274              | RFM-21       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.2           | 10.4                  | 32.8            | -0.5              | -1.6        | 32111          | 9.4        | 59      | 15.0    | 65      | 18.3    | 10.7       | 0.7        | 11.7         | 34.8          | 70           | 21.1         | 86             | 30.1           | 31.4                  | 99.1            | -1.5              | -4.7        | 98923          | 28.4       | 110     | 43.3    | 90      | 32.2    | 9.7        | 0.6   | 9.0        | 26.8         | 5/8" hePEX    | 6"   | 152          | 12       | 537         | 164        | 1.5 |
| RFHC-Zone 10-4   | 1687              | 157              | RFM-22       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 67             | 19.2           | 10.4                  | 32.8            | -0.5              | -1.6        | 18380          | 5.4        | 59      | 15.0    | 65      | 18.3    | 6.1        | 0.4        | 3.1          | 9.2           | 70           | 21.1         | 86             | 30.1           | 31.4                  | 99.1            | -1.5              | -4.7        | 55502          | 16.3       | 110     | 43.3    | 90      | 32.2    | 5.6        | 0.4   | 2.4        | 7.1          | 5/8" hePEX    | 6"   | 152          | 12       | 327         | 100        | 1.5 |
| RFHC-Zone 11     | 1049              | 97               | RFM-23       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 69             | 20.4           | 7.8                   | 24.6            | -9.3              | -29.3       | 17938          | 5.3        | 59      | 15.0    | 65      | 18.3    | 6.0        | 0.4        | 6.1          | 18.2          | 70           | 21.1         | 83             | 28.2           | 24.9                  | 78.6            | -15.8             | -49.8       | 42694          | 12.5       | 110     | 43.3    | 90      | 32.2    | 4.3        | 0.3   | 3.1        | 9.1          | 5/8" hePEX    | 6"   | 152          | 7        | 326         | 99         | 3.6 |
| RFHC-Zone 12     | 765               | 71               | RFM-24       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 72             | 22.0           | 4.2                   | 13.3            | -9.9              | -31.2       | 10787          | 3.2        | 59      | 15.0    | 65      | 18.3    | 3.6        | 0.2        | 6.4          | 19.2          | 70           | 21.1         | 76             | 24.5           | 11.9                  | 37.5            | -17.5             | -55.2       | 22491          | 6.6        | 110     | 43.3    | 90      | 32.2    | 2.2        | 0.1   | 2.5        | 7.6          | 5/8" hePEX    | 6"   | 152          | 4        | 339         | 103        | 3.7 |
| RFHC-Zone 13     | 597               | 55               | RFM-25       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 72             | 22.0           | 4.2                   | 13.3            | -9.9              | -31.2       | 8418           | 2.5        | 59      | 15.0    | 65      | 18.3    | 2.8        | 0.2        | 4.4          | 13.1          | 70           | 21.1         | 76             | 24.5           | 11.9                  | 37.5            | -17.5             | -55.2       | 17257          | 4.8        | 110     | 43.3    | 90      | 32.2    | 1.8        | 0.1   | 1.7        | 5.2          | 5/8" hePEX    | 6"   | 152          | 3        | 302         | 100        | 3.7 |
| RFHC-Zone 14     | 626               | 58               | RFM-26       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 72             | 22.0           | 4.2                   | 13.3            | -9.9              | -31.2       | 8827           | 2.6        | 59      | 15.0    | 65      | 18.3    | 2.9        | 0.2        | 5.8          | 17.4          | 70           | 21.1         | 76             | 24.5           | 11.9                  | 37.5            | -17.5             | -55.2       | 18404          | 5.4        | 110     | 43.3    | 90      | 32.2    | 1.8        | 0.1   | 2.3        | 6.9          | 5/8" hePEX    | 6"   | 152          | 4        | 458         | 140        | 3.7 |
| RFHC-Zone 15     | 301               | 28               | RFM-27       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 69             | 20.4           | 7.8                   | 24.6            | -9.3              | -29.3       | 5147           | 1.5        | 59      | 15.0    | 65      | 18.3    | 1.7        | 0.1        | 6.2          | 18.5          | 70           | 21.1         | 83             | 28.2           | 24.9                  | 78.6            | -15.8             | -49.8       | 42251          | 3.6        | 110     | 43.3    | 90      | 32.2    | 1.2        | 0.1   | 3.1        | 9.3          | 5/8" hePEX    | 6"   | 152          | 2        | 365         | 111        | 3.6 |
| RFHC-Zone 16     | 289               | 26               | RFM-28       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 72             | 22.0           | 4.2                   | 13.3            | -9.9              | -31.2       | 4628           | 1.3        | 59      | 15.0    | 65      | 18.3    | 1.7        | 0.1        | 5.2          | 12.3          | 70           | 21.1         | 76             | 24.5           | 11.9                  | 37.5            | -17.5             | -55.2       | 16933          | 4.0        | 110     | 43.3    | 90      | 32.2    | 1.2        | 0.1   | 2.4        | 7.2          | 5/8" hePEX    | 6"   | 152          | 2        | 324         | 100        | 3.7 |
| RFHC-Zone 17     | 124               | 12               | RFM-29       | Uponor       | Stainless Steel | 0        | 75           | 23.8         | 72             | 22.0           | 4.2                   | 13.3            | -9.9              | -31.2       | 1748           | 0.         |         |         |         |         |            |            |              |               |              |              |                |                |                       |                 |                   |             |                |            |         |         |         |         |            |       |            |              |               |      |              |          |             |            |     |

DISCLAIMER:  
Note: Details of the tubing bend radius have been simplified for clarity. Consult tubing manufacturer for specific bend radius recommendations.

PROJECT:  
**Dockstader PRPS**

CUSTOMER:

PROJECT NO.:  
**54906H S R1**

SCALE:  
**1/8"=1'**

DRAWING NAME:  
**Snow Melt**

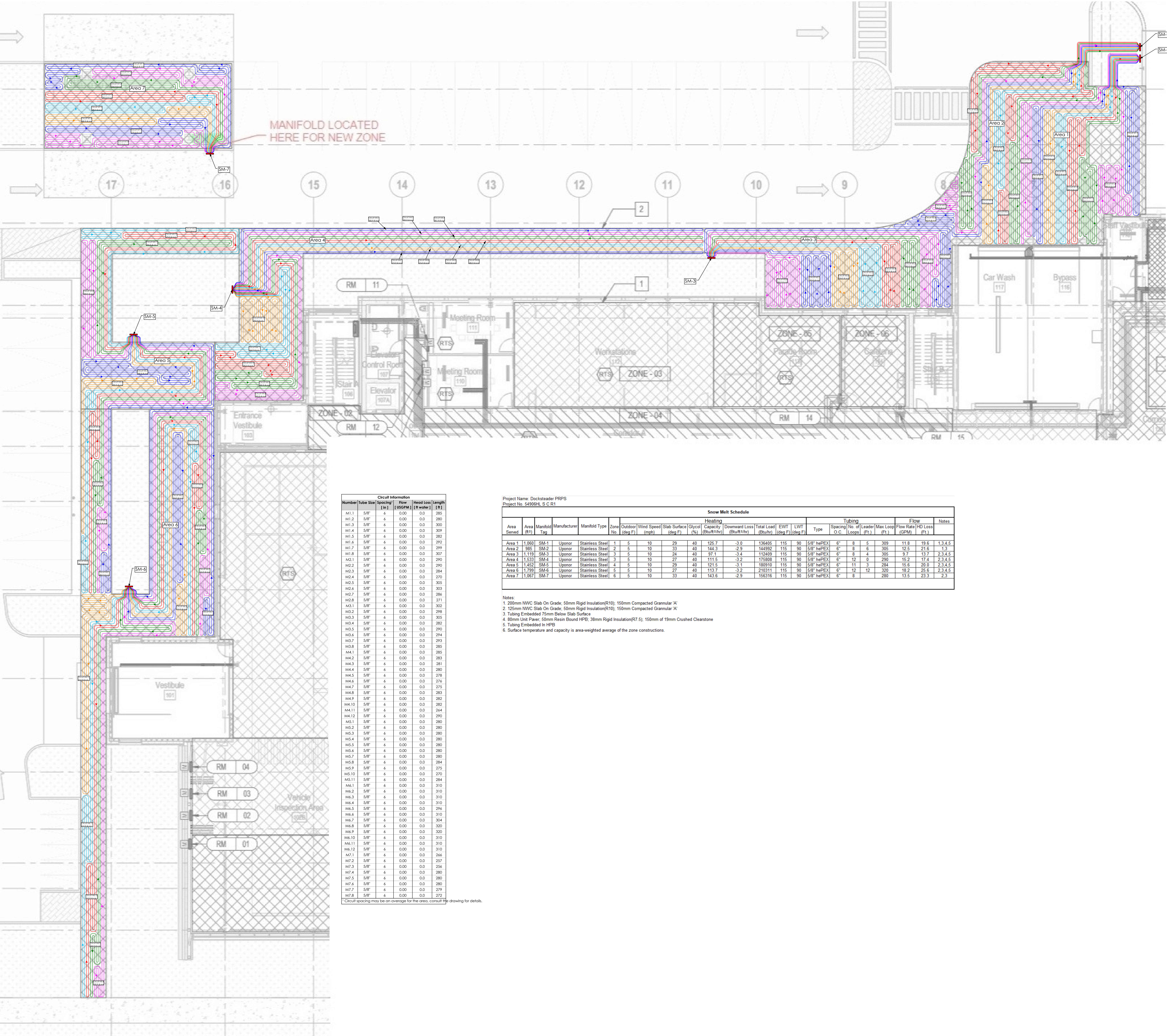
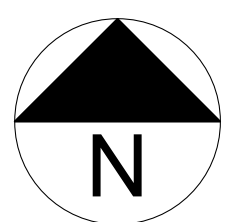
DRAWN BY:  
**Tracy Wang**

Created Using LoopCAD 2026 26.0.0095 (2026-09-03)

DATE:  
**2026-09-03**

REVISIONS

| No | Desc                                           | Date       |
|----|------------------------------------------------|------------|
| 1  | Updated site plan; adjusted manifold locations | 2026-09-03 |



# Wirsbo hePEX coils

## Project information

|                                |                  |                 |                |
|--------------------------------|------------------|-----------------|----------------|
| Job name:                      | Dockstader PRPS  | Location:       | Peel Region    |
| Engineer:                      | Introba          | Date submitted: | 09/10/2026     |
| Contractor:                    | MSB Mechanical   | Submitted by:   | Craig Stoneman |
| Manufacturer's representative: | The Morgan Group | Approved by:    |                |

## Technical data

|                       |                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------|
| Material              | PEX                                                                                                            |
| Temp/pressure ratings | 73 °F (23 °C) at 160 psi (11 bar)<br>180 °F (82 °C) at 100 psi (6.9 bar)<br>200 °F (93 °C) at 80 psi (5.5 bar) |

## Product information and application use

Wirsbo hePEX™ tubing is used for closed-loop heating and cooling applications, including radiant and hydronic distribution. It has an EVOH oxygen diffusion barrier that meets German DIN 4726/9.

Note: Temperature and pressure ratings stated are hydrostatic ratings. For domestic hot-water (DHW) and DHW recirculation installations, operating conditions should not exceed 140°F (60°C) at 80 psi (5.5 bar) in accordance with ASTM F2023. For additional information regarding application-specific temperature and pressure ratings, refer to the Uponor PEX Piping Systems Design and Installation Manual.



| Part name                           | Part no. | Inner diameter coil [inch] | Outer diameter coil [inch] | Coil width [inch] | Inner diameter pipe [inch] | Outer diameter pipe [inch] | Weight per UOM [lbs/UOM] |
|-------------------------------------|----------|----------------------------|----------------------------|-------------------|----------------------------|----------------------------|--------------------------|
| 5/16" Wirsbo hePEX, 100-ft. coil    | A1140313 | 10.5                       | 16.5                       | 3.5               | 0.292                      | 0.43                       | 3                        |
| 5/16" Wirsbo hePEX, 250-ft. coil    | A1180313 | 10.5                       | 24.5                       | 4.5               | 0.292                      | 0.43                       | 7.49                     |
| 5/16" Wirsbo hePEX, 1,000-ft. coil  | A1220313 | 15.75                      | 32                         | 6.5               | 0.292                      | 0.43                       | 29.98                    |
| 3/8" Wirsbo hePEX, 100-ft. coil     | A1140375 | 15.75                      | 16.5                       | 3.5               | 0.35                       | 0.5                        | 4.19                     |
| 3/8" Wirsbo hePEX, 400-ft. coil     | A1210375 | 15.75                      | 32                         | 4.5               | 0.35                       | 0.5                        | 16.76                    |
| 1/2" Wirsbo hePEX, 100-ft. coil     | A1140500 | 15.75                      | 23.5                       | 3.5               | 0.475                      | 0.625                      | 5.3                      |
| 1/2" Wirsbo hePEX, 300-ft. coil     | A1250500 | 19.7                       | 32                         | 4                 | 0.475                      | 0.625                      | 22.79                    |
| 1/2" Wirsbo hePEX, 500-ft. coil     | A1260500 | 15.75                      | 32                         | 6.5               | 0.475                      | 0.625                      | 26.48                    |
| 1/2" Wirsbo hePEX, 1,000-ft. coil   | A1220500 | 15.75                      | 32                         | 10.5              | 0.475                      | 0.625                      | 52.96                    |
| 5/8" Wirsbo hePEX, 100-ft. coil     | A1140625 | 15.75                      | 24                         | 5                 | 0.574                      | 0.75                       | 7.6                      |
| ✓ 5/8" Wirsbo hePEX, 300-ft. coil   | A1250625 | 15.75                      | 32                         | 6.5               | 0.574                      | 0.75                       | 30.07                    |
| ✓ 5/8" Wirsbo hePEX, 400-ft. coil   | A1210625 | 15.75                      | 32                         | 6.5               | 0.574                      | 0.75                       | 30.39                    |
| ✓ 5/8" Wirsbo hePEX, 1,000-ft. coil | A1220625 | 15.75                      | 32                         | 16                | 0.574                      | 0.75                       | 75.98                    |
| 3/4" Wirsbo hePEX, 100-ft. coil     | A1140750 | 15.75                      | 32                         | 3                 | 0.671                      | 0.875                      | 10.02                    |
| 3/4" Wirsbo hePEX, 300-ft. coil     | A1250750 | 22                         | 32                         | 8                 | 0.671                      | 0.875                      | 51.29                    |
| 3/4" Wirsbo hePEX, 500-ft. coil     | A1240750 | 15.75                      | 32                         | 10.5              | 0.671                      | 0.875                      | 85.48                    |
| 3/4" Wirsbo hePEX, 1,000-ft. coil   | A1220750 | 15.75                      | 32                         | 16                | 0.671                      | 0.875                      | 100.22                   |
| 1" Wirsbo hePEX, 100-ft. coil       | A1141000 | 22.7                       | 32                         | 6.5               | 0.862                      | 1.125                      | 17.1                     |
| 1" Wirsbo hePEX, 300-ft. coil       | A1251000 | 22.7                       | 32                         | 16                | 0.862                      | 1.125                      | 72.29                    |
| 1" Wirsbo hePEX, 500-ft. coil       | A1241000 | 22.7                       | 40                         | 13.8              | 0.862                      | 1.125                      | 15.89                    |
| 1 1/4" Wirsbo hePEX, 100-ft. coil   | A1141250 | 27                         | 45                         | 6                 | 1.054                      | 1.375                      | 24.1                     |
| 1 1/4" Wirsbo hePEX, 300-ft. coil   | A1251250 | 22.7                       | 40                         | 13.8              | 1.054                      | 1.375                      | 104.79                   |
| 1 1/2" Wirsbo hePEX, 100-ft. coil   | A1141500 | 27                         | 45                         | 6                 | 1.244                      | 1.625                      | 34.93                    |
| 1 1/2" Wirsbo hePEX, 300-ft. coil   | A1251500 | 27                         | 45                         | 19                | 1.244                      | 1.625                      | 180.29                   |
| 2" Wirsbo hePEX, 100-ft. coil       | A1142000 | 27                         | 44                         | 9                 | 1.629                      | 2.125                      | 60.1                     |
| 2" Wirsbo hePEX, 300-ft. coil       | A1252000 | 27                         | 51                         | 15                | 1.629                      | 2.125                      | 276.56                   |
| 3/8" Wirsbo hePEX, 1,000-ft. coil   | A1220375 | 15.75                      | 32                         | 6.5               | 0.35                       | 0.5                        | 41.91                    |



| Part name          | Part no. | Codes                                                         | Standards                                                                                                                                                  | Listings                                                                           |
|--------------------|----------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Wirsbo hePEX coils | All      | IMC UPC IBC IRC NBC of Canada IPC UFGS NPC of Canada UMC NSPC | ASTM F1960 ASTM E119/UL 263 ASTM F876 ASTM E84 ASTM F2023 ASTM E814/ULC S115 ASTM F877 CSA B137.5 ULC S102.2 CSA B214 NSF/ANSI/CAN 61 ULC S101 NSF/ANSI 14 | IAPMO-ES ICC-ES-PMG cNSFus-rfh cNSFus-pw CCMC PPI TR-4 CSA UL U.P.Code cQAlus P321 |

| Installation                                                                                                                                                                    | Related applications                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approved fitting system is ProPEX®. 1 Refer to the Uponor Complete Design Assistance Manual (CDAM) or Hydronic Piping Design Assistance Manual (HPDAM) for further information. | Radiant Heating and Cooling Systems<br>Permafrost Protection Systems<br>Turf Conditioning Systems<br>Chilled water distribution<br>Hot water distribution |

| Footnotes | Contact information                                                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -         | Uponor Inc.<br>5925 148th Street West<br>Apple Valley, MN 55124<br>T 800.321.4739<br>F 952.891.2008<br>Uponor Ltd.<br>6510 Kennedy Road<br>Mississauga, ON L5T 2X4<br>T 888.594.7726<br>F 800.638.9517 |

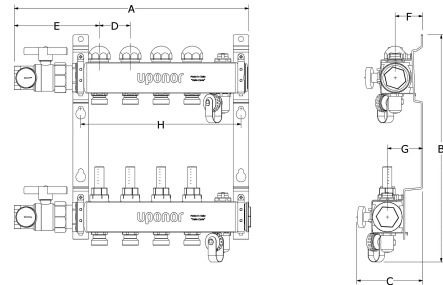
# Stainless-steel manifold assembly, 1 1/4" with flow meter

## Project information

|                                |                  |                 |                |
|--------------------------------|------------------|-----------------|----------------|
| Job name:                      | Dockstader PRPS  | Location:       | Peel Region    |
| Engineer:                      | Introba          | Date submitted: | 09/10/2026     |
| Contractor:                    | MSB Mechanical   | Submitted by:   | Craig Stoneman |
| Manufacturer's representative: | The Morgan Group | Approved by:    |                |

## Technical data

|                         |                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------|
| Material                | Stainless Steel                                                                                           |
| Subcomponent Material   | Manifold Components:<br>Brass CW614N-UNI EN12164, CW617N-UNI EN12165                                      |
| Manifold size           | 1.25 inch                                                                                                 |
| Loop Cv                 | 1.1 Cv                                                                                                    |
| End type 1              | ISO 228-G 1-1/4"                                                                                          |
| End type 2              | ISO 228-G 3/4"                                                                                            |
| Temp/pressure ratings   | 68 °F (20 °C) at 145 psi (10 bar)<br>158 °F (70 °C) at 87 psi (6 bar)<br>194 °F (90 °C) at 44 psi (3 bar) |
| Max. fluid flow rate    | 21 gpm                                                                                                    |
| Prop 65 label required? | Yes                                                                                                       |



## Product information and application use

The 1 1/4" Stainless-steel Manifold Assembly with flow meter comes fully assembled and ready for installation. Use the included flow meters located on the supply manifold to balance the loops across the manifold. The return manifold comes equipped with isolation valves for easy loop isolation. For multiple zoning on the manifold, the black caps are removed and replaced with Uponor Thermal Actuators. The manifold body ends have R32 union connections and the loop outlets have R20 male threads. The included manifold supply and return ball valves come with an R32 x 1" NPT or R32 x 1 1/4" NPT connection. Note: The 1 1/4" manifolds with 2 through 7 loops are strategically designed with 1" NPT ball valves to avoid adding extra costs while maintaining the maximum flow of the manifold assembly.

## Snowmelt Manifolds

| Part name                                                                              | Part no. | A [inch] | B [inch] | C [inch] | D [inch] | E [inch] | F [inch] | G [inch] | H [inch] | Weight per UOM [lbs/UOM] |
|----------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--------------------------|
| Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 2 loops    | A2720202 | 10.933   | 14.46    | 3.542    | 1.969    | 5.425    | 1.75     | 2.25     | 6.342    | 9.22                     |
| Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 3 loops    | A2720302 | 12.902   | 14.46    | 3.542    | 1.969    | 5.425    | 1.75     | 2.25     | 8.249    | 10.12                    |
| Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 4 loops    | A2720402 | 14.87    | 14.46    | 3.542    | 1.969    | 5.425    | 1.75     | 2.25     | 10.211   | 11.02                    |
| Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 5 loops    | A2720502 | 16.839   | 14.46    | 3.541    | 1.969    | 5.425    | 1.75     | 2.25     | 12.131   | 11.92                    |
| Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 6 loops    | A2720602 | 19.066   | 14.47    | 3.456    | 1.969    | 5.504    | 1.75     | 2.25     | 14.175   | 12.82                    |
| Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 7 loops    | A2720702 | 20.776   | 14.46    | 3.542    | 1.969    | 5.425    | 1.75     | 2.25     | 16.156   | 13.72                    |
| ✓ Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 8 loops  | A2720802 | 23.095   | 14.46    | 3.439    | 1.969    | 5.776    | 1.75     | 2.25     | 18.262   | 15.32                    |
| Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 10 loops   | A2721002 | 27.291   | 14.47    | 3.415    | 1.969    | 5.855    | 1.75     | 2.25     | 22.189   | 17.12                    |
| ✓ Stainless-steel Manifold Assembly, 1 1/4" with flow meter, B&I, ball valve, 12 loops | A2721202 | 31.228   | 14.47    | 3.415    | 1.969    | 5.855    | 1.75     | 2.25     | 26.106   | 18.92                    |

## Installation

The 1 1/4" Stainless-steel Manifolds are fully assembled and ready for installation. Manifold assembly comes with mounting brackets, isolation valves, balancing valves, fill/purge ports, air vents, ball valves, temperature gauges and end caps. Manifold adapters and fittings are sold separately. For further information, refer to the Uponor Stainless-steel Manifold Installation Guide.

## Related applications

Radiant Heating and Cooling Systems  
Permafrost Protection Systems  
Turf Conditioning Systems  
AquaSAFE™ Fire Safety Systems

## Footnotes

-

## Contact information

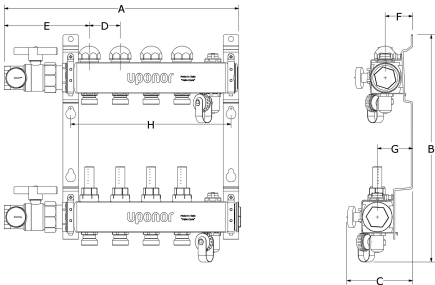
Uponor Inc.  
5925 148th Street West  
Apple Valley, MN 55124  
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Uponor Ltd.  
6510 Kennedy Road  
Mississauga, ON L5T 2X4  
T 888.594.7726  
F 800.638.9517

Stainless-steel manifold assembly, 1" with flow meter

| Project information            |                  |                 |                |
|--------------------------------|------------------|-----------------|----------------|
| Job name:                      | Dockstader PRPS  | Location:       | Peel Region    |
| Engineer:                      | Introba          | Date submitted: | 09/102026      |
| Contractor:                    | MSB Mechanical   | Submitted by:   | Craig Stoneman |
| Manufacturer's representative: | The Morgan Group | Approved by:    |                |

| Technical data          |                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------|
| Material                | Stainless Steel                                                                                           |
| Subcomponent Material   | Manifold Components:<br>Brass CW614N-UNI EN12164, CW617N-UNI EN12165                                      |
| Manifold size           | 1 inch                                                                                                    |
| Loop Cv                 | 1.1 Cv                                                                                                    |
| End type 1              | ISO 228-G 1-1/4"                                                                                          |
| End type 2              | ISO 228-G 3/4"                                                                                            |
| Temp/pressure ratings   | 68 °F (20 °C) at 145 psi (10 bar)<br>158 °F (70 °C) at 87 psi (6 bar)<br>194 °F (90 °C) at 44 psi (3 bar) |
| Max. fluid flow rate    | 14 gpm                                                                                                    |
| Prop 65 label required? | Yes                                                                                                       |



Product information and application use

The 1" Stainless-steel Manifold Assembly with flow meter comes fully assembled and ready for installation. Use the included flow meters located on the supply manifold to balance the loops across the manifold. The return manifold comes equipped with isolation valves for easy loop isolation. For multiple zoning on the manifold, the black caps are removed and replaced with Uponor Thermal Actuators. The manifold body ends have R32 union connections and the loop outlets have R20 male threads. The included manifold supply and return ball valves come with an R32 x 1" NPT connection.

Radiant Floor Heating Manifolds

|   | Part name                                                                        | Part no. | A [inch] | B [inch] | C [inch] | D [inch] | E [inch] | F [inch] | G [inch] | H [inch] | Weight per UOM [lbs/UOM] |
|---|----------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|--------------------------|
| ✓ | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 2 loops  | A2700202 | 11.16    | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 6.146    | 9.2                      |
| ✓ | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 3 loops  | A2700302 | 13.128   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 8.273    | 10.4                     |
| ✓ | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 4 loops  | A2700402 | 15.097   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 10.2     | 11.16                    |
|   | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 5 loops  | A2700502 | 17.065   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 12.198   | 12.2                     |
|   | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 6 loops  | A2700602 | 19.034   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 14.226   | 13.25                    |
| ✓ | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 7 loops  | A2700702 | 21.002   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 16.149   | 12.22                    |
| ✓ | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 8 loops  | A2700802 | 22.971   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 18.144   | 13.12                    |
| ✓ | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 10 loops | A2701002 | 26.908   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 22.032   | 14.92                    |
| ✓ | Stainless-steel Manifold Assembly, 1" with flow meter, B&I, ball valve, 12 loops | A2701202 | 30.845   | 14.119   | 3.842    | 1.969    | 5.488    | 1.55     | 2.05     | 25.994   | 16.72                    |

| Installation                                                                                                                                                                                                                                                                                                                                                                                  | Related applications                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| The 1" Stainless-steel Manifolds are fully assembled and ready for installation. Manifold assembly comes with mounting brackets, isolation valves, balancing valves, fill/purge ports, air vents, ball valves, temperature gauges and end caps. Manifold adapters and fittings are sold separately. For further information, refer to the Uponor Stainless-steel Manifold Installation Guide. | Radiant Heating and Cooling Systems<br>Permafrost Protection Systems<br>Turf Conditioning Systems<br>Hydronic Heating and Cooling Systems |

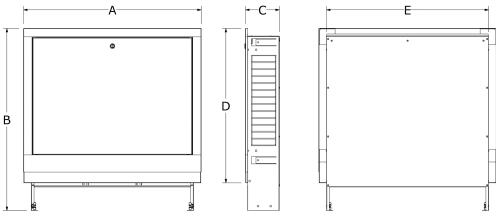
| Footnotes | Contact information                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -         | Uponor Inc.<br>5925 148th Street West<br>Apple Valley, MN 55124<br>T 800.321.4739<br>F 952.891.2008<br><br>Uponor Ltd.<br>6510 Kennedy Road<br>Mississauga, ON L5T 2X4<br>T 888.594.7726<br>F 800.638.9517 |

Manifold wall cabinets, recessed mount

| Project information            |                  |                 |                |
|--------------------------------|------------------|-----------------|----------------|
| Job name:                      | Dockstader PRPS  | Location:       | Peel Region    |
| Engineer:                      | Introba          | Date submitted: | 04/23/2026     |
| Contractor:                    | MSB Mechanical   | Submitted by:   | Craig Stoneman |
| Manufacturer's representative: | The Morgan Group | Approved by:    |                |

| Technical data                          |       |
|-----------------------------------------|-------|
| Material                                | Metal |
| Prop 65 label required?                 | No    |
| Product information and application use |       |

Uponor Manifold Wall Cabinets, Recessed Mount, are designed to be fitted into a 2x4 or 2x6 wall or opening. The cabinets accommodate Uponor Stainless-Steel Manifolds, Uponor Engineered Polymer (EP) Manifolds, or Uponor Commercial Stainless-Steel Manifolds for use in radiant heating and/or cooling, snow and ice melting, turf conditioning, and permafrost prevention (cold storage) applications.



Radiant Floor Heating Cabinets

|   | Part name                                                     | Part no. | A [inch] | B [inch] | C [inch] | D [inch] | E [inch] | Weight per UOM [lbs/UOM] |
|---|---------------------------------------------------------------|----------|----------|----------|----------|----------|----------|--------------------------|
| ✓ | Manifold Wall Cabinet, Recessed Mount, 28" W x 31" H x 3.5" D | A2603128 | 30.197   | 30.906   | 3.937    | 29.587   | 27.559   | 19.75                    |
| ✓ | Manifold Wall Cabinet, Recessed Mount, 42" W x 31" H x 3.5" D | A2603142 | 42.008   | 30.906   | 3.937    | 29.587   | 39.37    | 26.37                    |
|   | Manifold Wall Cabinet, Recessed Mount, 30" W x 31" H x 5.4" D | A2613130 | 30.197   | 30.906   | 5.78     | 29.587   | 27.559   | 19.75                    |
|   | Manifold Wall Cabinet, Recessed Mount, 56" W x 31" H x 5.4" D | A2613156 | 55.638   | 30.906   | 5.78     | 29.587   | 53       | 35.19                    |

| Installation                                                                                                                                                                                                                                | Related applications                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Using the included hardware, mount the cabinet in either a 2x4 or 2x6 stud wall and attach the door-trim package. For detailed installation instructions, refer to the Uponor Recessed-Mount Wall Cabinets Instruction Sheet on uponor.com. | Radiant Heating and Cooling Systems<br>Snow and Ice Melting Systems<br>Turf Conditioning Systems<br>Permafrost Prevention (Cold Storage) Systems |

| Notes |
|-------|
|-------|

White color RAL9016

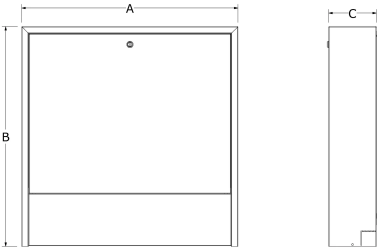
| Footnotes | Contact information                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -         | Uponor Inc.<br>5925 148th Street West<br>Apple Valley, MN 55124<br>T 800.321.4739<br>F 952.891.2008<br><br>Uponor Ltd.<br>6510 Kennedy Road<br>Mississauga, ON L5T 2X4<br>T 888.594.7726<br>F 800.638.9517 |

Manifold wall cabinets, surface mount

| Project information            |                  |                 |                |
|--------------------------------|------------------|-----------------|----------------|
| Job name:                      | Dockstader PRPS  | Location:       | Peel Region    |
| Engineer:                      | Introba          | Date submitted: | 04/23/2026     |
| Contractor:                    | MSB Mechanical   | Submitted by:   | Craig Stoneman |
| Manufacturer's representative: | The Morgan Group | Approved by:    |                |

| Technical data                          |       |
|-----------------------------------------|-------|
| Material                                | Metal |
| Prop 65 label required?                 | No    |
| Product information and application use |       |

Uponor Manifold Wall Cabinets, Surface Mount, are designed to be mounted directly on various surfaces, including concrete block and stud walls. They feature a separable back plate for easy installation and mounting. The cabinets accommodate Uponor Stainless-Steel Manifolds, Uponor Engineered Polymer (EP) Manifolds, or Uponor Commercial Stainless-Steel Manifolds for use in radiant heating and/or cooling, snow and ice melting, turf conditioning, and permafrost prevention (cold storage) applications.



Radiant Floor Heating Cabinets

|   | Part name                                                    | Part no. | A [Inch] | B [Inch] | C [Inch] | Weight per UOM [lbs/UOM] |
|---|--------------------------------------------------------------|----------|----------|----------|----------|--------------------------|
| ✓ | Manifold Wall Cabinet, Surface Mount, 28" W x 28" H x 6.1" D | A2702828 | 27.559   | 27.953   | 6.106    | 18.65                    |
| ✓ | Manifold Wall Cabinet, Surface Mount, 53" W x 28" H x 6.1" D | A2702853 | 53       | 27.953   | 6.106    | 31.88                    |

| Installation                                                                                                                                                                                                                        | Related applications                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Using the included hardware, mount the cabinet on the desired surface, then attach the enclosure and door. For detailed installation instructions, refer to the Uponor Surface-Mount Wall Cabinets Instruction Sheet on uponor.com. | Radiant Heating and Cooling Systems<br>Snow and Ice Melting Systems<br>Turf Conditioning Systems<br>Permafrost Prevention (Cold Storage) Systems |

| Notes |
|-------|
|-------|

White color RAL9016

| Footnotes | Contact information                                                                                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -         | Uponor Inc.<br>5925 148th Street West<br>Apple Valley, MN 55124<br>T 800.321.4739<br>F 952.891.2008<br>Uponor Ltd.<br>6510 Kennedy Road<br>Mississauga, ON L5T 2X4<br>T 888.594.7726<br>F 800.638.9517 |

ProPEX manifold fitting assemblies

Project information

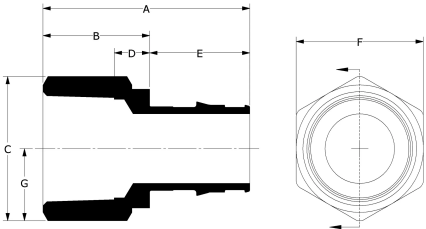
|                                |                  |                 |                |
|--------------------------------|------------------|-----------------|----------------|
| Job name:                      | Dockstader PRPS  | Location:       | Peel Region    |
| Engineer:                      | Introba          | Date submitted: | 04/23/2026     |
| Contractor:                    | MSB Mechanical   | Submitted by:   | Craig Stoneman |
| Manufacturer's representative: | The Morgan Group | Approved by:    |                |

Technical data

|                         |       |
|-------------------------|-------|
| Material                | Brass |
| Prop 65 label required? | Yes   |

Product information and application use

The Uponor ProPEX® Fitting Assembly threads onto Uponor manifold loop outlets. The fitting features an R20 or R25 thread with a ProPEX connection to connect Uponor crosslinked polyethylene (PEX) tubing to the manifold in radiant heating and cooling applications.



| Part name                                                        | Part no. | A [inch] | B [inch] | C [inch] | D [inch] | E [inch] | F [inch] | G [inch] | End Type 1  | End Type 2     | Operating Temperature 1 [°F] |
|------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|-------------|----------------|------------------------------|
| 3/8" ProPEX Fitting Assembly, R20 Thread                         | Q4020375 | 1.196    | 0.605    | 1.368    | 0.285    | 0.591    | 1.187    | 0.684    | ProPEX 3/8" | ISO 228-G 3/4" | 73                           |
| 1/2" ProPEX Fitting Assembly, R20 Thread                         | Q4020500 | 1.321    | 0.612    | 1.368    | 0.285    | 0.709    | 1.187    | 0.684    | ProPEX 1/2" | ISO 228-G 3/4" | -                            |
| ✓ 5/8" ProPEX Fitting Assembly, R20 Thread                       | Q4020625 | 1.468    | 0.602    | 1.368    | 0.285    | 0.866    | 1.187    | 0.684    | ProPEX 5/8" | ISO 228-G 3/4" | -                            |
| 3/4" ProPEX Fitting Assembly, R20 Thread                         | Q4020750 | 1.55     | 0.605    | 1.368    | 0.285    | 0.945    | 1.187    | 0.684    | -           | -              | -                            |
| 5/8" ProPEX Fitting Assembly for Commercial Manifold, R25 thread | Q4050625 | 1.634    | 0.768    | 1.729    | 0.275    | 0.866    | 1.5      | -        | ProPEX 5/8" | ISO 228-G 1"   | 73                           |
| 3/4" ProPEX Fitting Assembly for Commercial Manifold, R25 thread | Q4050750 | 1.713    | 0.768    | 1.729    | 0.275    | 0.945    | 1.5      | -        | ProPEX 3/4" | ISO 228-G 1"   | 73                           |
| 1" ProPEX Fitting Assembly for Commercial Manifold, R25 thread   | Q4051000 | 1.949    | 0.768    | 1.729    | 0.275    | 1.181    | 1.5      | -        | ProPEX 1"   | ISO 228-G 1"   | 73                           |

| Part name                                                                                 | Part no. | Operating Temperature 2 [°F] | Operating Temperature 3 [°F] | Cv    | Equivalent length through [ft] | Weight per UOM [lbs/UOM] |
|-------------------------------------------------------------------------------------------|----------|------------------------------|------------------------------|-------|--------------------------------|--------------------------|
| <input type="checkbox"/> 3/8" ProPEX Fitting Assembly, R20 Thread                         | Q4020375 | 180                          | 200                          | -     | -                              | 0.18                     |
| <input type="checkbox"/> 1/2" ProPEX Fitting Assembly, R20 Thread                         | Q4020500 | -                            | -                            | -     | -                              | 0.18                     |
| ✓ 5/8" ProPEX Fitting Assembly, R20 Thread                                                | Q4020625 | -                            | -                            | -     | -                              | 0.2                      |
| <input type="checkbox"/> 3/4" ProPEX Fitting Assembly, R20 Thread                         | Q4020750 | -                            | -                            | -     | -                              | 0.19                     |
| <input type="checkbox"/> 5/8" ProPEX Fitting Assembly for Commercial Manifold, R25 thread | Q4050625 | 180                          | 200                          | 7.75  | 4.13                           | 0.37                     |
| <input type="checkbox"/> 3/4" ProPEX Fitting Assembly for Commercial Manifold, R25 thread | Q4050750 | 180                          | 200                          | 12.07 | 3.18                           | 0.36                     |
| <input type="checkbox"/> 1" ProPEX Fitting Assembly for Commercial Manifold, R25 thread   | Q4051000 | 180                          | 200                          | 23    | 3.22                           | 0.38                     |

| Part name                                                                                                                                                                                                                                                                                                                  | Part no. | Codes                                                                                               | Standards                       | Listings                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------|---------------------------------|----------------------------|
| ProPEX manifold fitting assemblies                                                                                                                                                                                                                                                                                         | All      | IMC IRC NBC of Canada UMC                                                                           | ASTM F1960 CSA B137.5 ASTM F877 | cNSFus-rfh ICC-ES-PMG 1012 |
| In addition, the following parts have additional codes, standards, or listings:                                                                                                                                                                                                                                            |          |                                                                                                     |                                 |                            |
| 3/8" ProPEX Fitting Assembly, R20 Thread                                                                                                                                                                                                                                                                                   | Q4020375 |                                                                                                     | R20: ISO 228-G ¾"               |                            |
| 5/8" ProPEX Fitting Assembly for Commercial Manifold, R25 thread                                                                                                                                                                                                                                                           | Q4050625 |                                                                                                     | R25: ISO 228-G 1" NSF/ANSI 14   | IAPMO-ES ICC PMG 1006      |
| 3/4" ProPEX Fitting Assembly for Commercial Manifold, R25 thread                                                                                                                                                                                                                                                           | Q4050750 |                                                                                                     | R25: ISO 228-G 1" NSF/ANSI 14   | IAPMO-ES ICC PMG 1006      |
| 1" ProPEX Fitting Assembly for Commercial Manifold, R25 thread                                                                                                                                                                                                                                                             | Q4051000 |                                                                                                     | R25: ISO 228-G 1" NSF/ANSI 14   | IAPMO-ES ICC PMG 1006      |
| Installation                                                                                                                                                                                                                                                                                                               |          | Related applications                                                                                |                                 |                            |
| R20 or R25 threads onto the Uponor manifold loop outlets. Compression end enables direct connection to Uponor PEX tubing. For more information, refer to the Uponor Radiant Floor Heating Installation Handbook. Accessories: Replacement O-ring for insert, R20 (A4021116); Replacement O-ring for insert, R25 (A4021000) |          | Radiant Heating and Cooling Systems                                                                 |                                 |                            |
| Footnotes                                                                                                                                                                                                                                                                                                                  |          | Contact information                                                                                 |                                 |                            |
| -                                                                                                                                                                                                                                                                                                                          |          | Uponor Inc.<br>5925 148th Street West<br>Apple Valley, MN 55124<br>T 800.321.4739<br>F 952.891.2008 |                                 |                            |
|                                                                                                                                                                                                                                                                                                                            |          | Uponor Ltd.<br>6510 Kennedy Road<br>Mississauga, ON L5T 2X4<br>T 888.594.7726<br>F 800.638.9517     |                                 |                            |

PVC bend supports

Project information

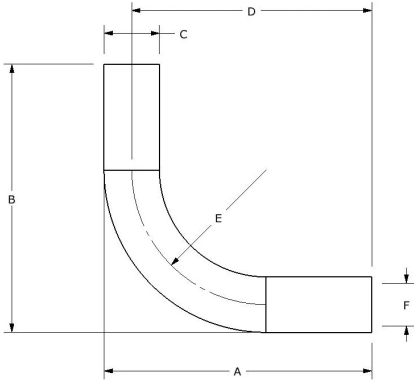
|                                |                  |                 |                |
|--------------------------------|------------------|-----------------|----------------|
| Job name:                      | Dockstader PRPS  | Location:       | Peel Region    |
| Engineer:                      | Introba          | Date submitted: | 04/23/2026     |
| Contractor:                    | MSB Mechanical   | Submitted by:   | Craig Stoneman |
| Manufacturer's representative: | The Morgan Group | Approved by:    |                |

Technical data

|                         |         |
|-------------------------|---------|
| Material                | Plastic |
| Subcomponent Material   | PVC     |
| Prop 65 label required? | Yes     |

Product information and application use

The Uponor PVC Bend Supports are for use in concrete slab installations. The sleeve is used to aid and support the PEX tubing as it enters/exits the concrete slab at a 90° angle.



| Part name                                         | Part no. | A [inch] | B [inch] | C [inch] | D [inch] | E [inch] | F [inch] | Weight per UOM [lbs/UOM] |
|---------------------------------------------------|----------|----------|----------|----------|----------|----------|----------|--------------------------|
| 3/4" PVC Elbow for 3/8" and 1/2" PEX Bend Support | A5500500 | 0.276    | 5.724    | 1.05     | 5.724    | 7.448    | 1.05     | 0.25                     |
| ✓ 1" PVC Elbow for 5/8" PEX Bend Support          | A5500625 | 6.384    | 6.384    | 1.32     | 5.724    | 7.448    | 1.04     | 0.4                      |
| 1 1/4" PVC Elbow for 3/4" PEX Bend Support        | A5500750 | 6.554    | 6.554    | 1.66     | 5.724    | 7.448    | 1.36     | 0.65                     |
| 1 1/2" PVC Elbow for 1" PEX Bend Support          | A5501000 | 6.674    | 6.674    | 1.9      | 5.724    | 7.448    | 1.58     | 0.75                     |

Installation

Feed the tubing through the bend sleeve at the 90° angle where the tubing enters the slab from the manifold. Refer to the Uponor Professional Plumbing Installation Guide or the Uponor Radiant Floor Installation Handbook for additional information.

Related applications

- Radiant Heating and Cooling Systems
- Permafrost Protection Systems
- Turf Conditioning Systems
- PEX-a Plumbing Systems

Footnotes

-

Contact information

Uponor Inc.  
5925 148th Street West  
Apple Valley, MN 55124  
T 800.321.4739  
F 952.891.2008

Uponor Ltd.  
6510 Kennedy Road  
Mississauga, ON L5T 2X4  
T 888.594.7726  
F 800.638.9517

# Snow Melt Control

- Operates a Motorized Valve, a Modulating Boiler, or a Multi-Boiler Control
- Melts the Snow based on the Slab Sensor or the Outdoor Sensor

# SMC



## Description:

The Heat-Timer SMC control system regulates the slab temperature to efficiently melt ice and snow. Therefore, it increases safety in many areas such as driveways, parking lots, staircases, walkways, station platforms, etc. It is designed to run an under slab hydronic heating system by controlling a motorized valve or a modulating boiler.

The SMC depends on the slab sensor to measure the slab temperature and the presence of precipitation. This sensor is designed to withstand heavy vehicle traffic. The sensor and its cabling must be installed before the slab concrete pour. In situations where the concrete slab is old and cannot be altered, the SMC can be used with an outdoor temperature sensor. The use of the outdoor temperature sensor eases the installation. However, it does not provide moisture detection that may increase energy usage.

In addition to the slab sensor, the SMC connects to two more temperature sensors to measure the slab supply and return temperatures. Their primary use is to help the SMC protect the slab from thermal stresses caused by rapid heating.

The SMC includes additional features that can be used to protect the plant and slab. The SMC can be used with a Boiler Return Sensor to help reduce the possibility of thermal shock and condensation to the boiler. Also, it is equipped with an Alarm output. This alarm is activated in many cases including sensor failure, the slab not receiving sufficient heat, or if the slab does not reach its melting temperature within 24 hours.

## Features:

- *Adjustable Warm-Weather-Cutoff, Cold-Weather-Cutoff, Idle Set Point, System Delta T ( $\Delta T$ ), Maximum Supply Temperature, Minimum Run-Time, Heating Curve, and Precipitation Sensitivity*

- *Operates a Motorized Valve, Modulating Boiler, System and Slab Pumps, and Alarm Output*
- *Display of Current Temperatures, Modes, and Alarm Status*
- *Boiler Return Protection (Sensor is optional)*

## Specifications

### SMC Specifications

|                                |                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------|
| Operating Temperature:         | 20°F/-6°C to 120°F/49°C                                                                       |
| Voltage Input:                 | 120 VAC 60 Hz                                                                                 |
| Maximum Input Rating:          | 48 VA Max                                                                                     |
| Output Built-in Relay Ratings: | 2 Amp inductive (Maximum of 1/4 HP) at 120 VAC 60 Hz                                          |
| Output Built-in Relays:        | 1 System, 1 Boiler, 1 Slab Pump, 1 Alarm, 2 Motorized Valve                                   |
| Output LEDs:                   | 1 System, 1 Boiler, 1 Slab Pump, 1 Alarm, 2 Motorized Valve, and 1 Power                      |
| Inputs:                        | Shutdown/Enable, Prove                                                                        |
|                                | 1 Slab Sensor (Composite), Outdoor, Slab Supply and Slab Return Temps, Boiler Return Temp     |
| Modulating Outputs             | 0-10V, 4-20mA, or Floating (3-Point)                                                          |
| Seasons:                       | Winter and Summer                                                                             |
| Operating Modes:               | Slab Sensor or Outdoor Sensor                                                                 |
| Control Mode:                  | Mixing Valve, Boiler, or External Interface (sends a 4-20mA set point to an external control) |
| Warm Weather Cutoff:           | 34°F/1°C to 44°F/7°C                                                                          |
| Cold Weather Cutoff:           | On, Off, and 0°F/18°C to 20°F/-7°C                                                            |
| Idle Set Point:                | Off, and 20°F/-7°C to 44°F/7°C                                                                |
| Minimum Runtime:               | 0 to 4 hours                                                                                  |
| Maximum Delay T:               | 10°F/6°C to 50°F/28°C                                                                         |
| Minimum Boiler Return:         | Off, and 40°F/4°C to 140°F/60°C                                                               |
| Heating Curve:                 | from -10 to +10                                                                               |
| Modulating Boiler Ignition %:  | from 1% to 50%                                                                                |
| Buttons:                       | 4 buttons each with multiple functions                                                        |
| Dimensions:                    | .7 1/8" W x 9 5/8" H x 3 1/2" D                                                               |
| Weight:                        | .2 pounds                                                                                     |

### Slab Sensor Specifications:

|                              |                            |
|------------------------------|----------------------------|
| Operating Temperature Range: | -30°F/-34°C to 250°F/121°C |
| Sensor Case                  | Brass                      |
| Cable length Provided        | 100 Feet                   |
| Weight:                      | 5 pound                    |

## SMC-Elite Items

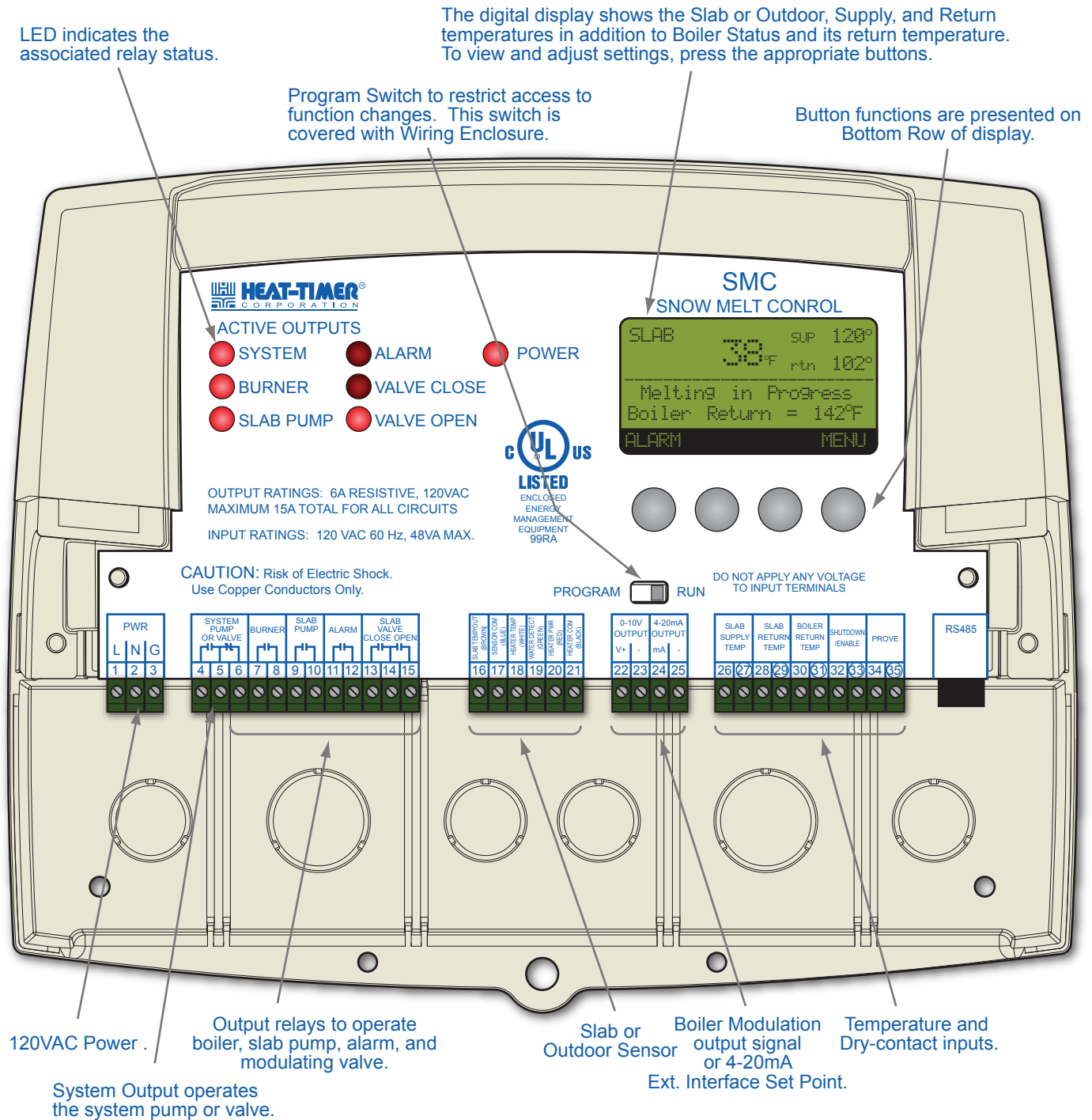
| Item Description                                                              | Part #    |
|-------------------------------------------------------------------------------|-----------|
| SMC Complete with Slab Sensor, Slab Supply, and Slab Return Sensors and Wells | 926820-00 |
| Slab Precipitation Sensor Complete with Sensor, Housing, and 100 ft of Cable  | 904200-00 |
| Slab Precipitation Sensor with only the Sensor and 100 ft of Cable            | 904198-00 |
| 3-in-1 Temperature Sensor (used as Boiler Return Sensor or Outdoor Sensor)    | 904220-00 |
| Well (for Boiler Return Sensor)                                               | 904011-00 |
| Vis-U-Larm                                                                    | 925011-00 |

HT# 056185-00B



20 New Dutch Lane, Fairfield, NJ 07004 973-575-4004 • Fax 973-575-4052 • <http://www.heat-timer.com>

# SMC Function Chart



HT# 056185-00B

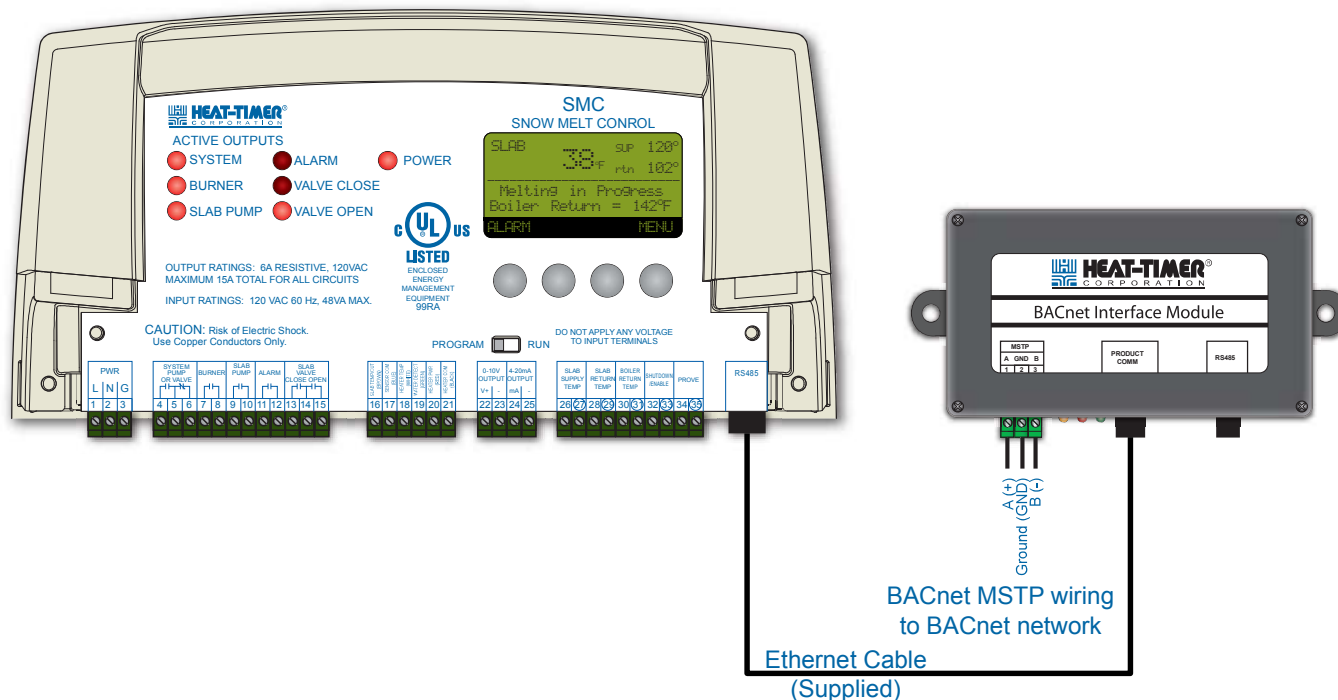


ISO 9001:2008  
CERTIFIED

20 New Dutch Lane, Fairfield, NJ 07004 973-575-4004 • Fax 973-575-4052 • <http://www.heat-timer.com>

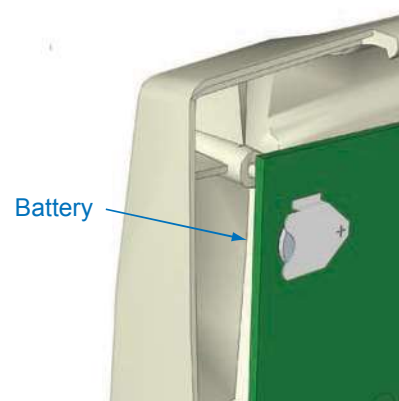
## WIRING TO BACNET MSTP

- For the SMC to communicate over the BACnet MSTP network, the use of the BACnet Interface Module is required. The BACnet Interface Module must be ordered separately (HT# 926781-00).
- Connect the provided CAT-5 cable to both the BACnet Interface Module (Product Comm Ethernet port) and the SMC's (RS485 port). The BACnet Interface Module is powered through the control. See BACnet Interface Module Manual for BACnet setup.



## ACTIVATE THE BATTERY

- The coin Lithium battery (CR2032) (HT# 020002-00) is on the back on the display module. It enables the control to maintain the clock time for up to 100 days of power outage. The clock is used in many aspects of the control operation including maintaining a 24-hour history of every sensor.
- To activate the battery, first disconnect the control power.
- Unscrew and remove the Enclosure Display Module to reveal the back of the control CPU Board.
- Remove the plastic tab that covers the battery to allow the battery holder contacts to touch the battery.



### **⚠️ ALERT**

Do not activate the battery unless you plan to keep the control continuously powered. If the control has no power, the battery will lose its charge in 100 days.

**Applications:**

The ESBE 3 and 4-Way rotary valves type F are designed for mixing or diverting applications on heating and cooling systems. The valves are capable of controlling and distributing water or non-corrosive liquids on closed looped systems to different zones by either manual operation or automatically via electronic motor actuator and controller.

**Features:**

- The F series of valves are a rotary design allowing for flexibility in field piping layout.
- Double mixing function for 4-Way valve, reducing the risk of condensation within heat source
- Valve position face plate is printed on both sides and can be rotated allowing a choice of valve mounting positions
- Easily adaptable with ESBE electronic motor actuator Series 90
- Suitable fluid of 100% water or glycol mixture (maximum 50% glycol) for heating or cooling closed loop systems.

**Ordering**  
**Information:**

3" Required for large zone

**3-Way Valve**

| Code No.        | Reference | Connection          | Cv* |
|-----------------|-----------|---------------------|-----|
| <b>065B8960</b> | 3 F 65-50 | 2-1/2" ANSI Flanged | 75  |
| <b>065B8961</b> | 3 F 65    | 2-1/2" ANSI Flanged | 105 |
| <b>065B8962</b> | 3 F 80    | 3" ANSI Flanged     | 175 |
| <b>065B8963</b> | 3 F 100   | 4" ANSI Flanged     | 265 |
| <b>065B8964</b> | 3 F 125   | 5" ANSI Flanged     | 325 |
| <b>065B8965</b> | 3 F 150   | 6" ANSI Flanged     | 465 |

**Note:**

ESBE 3-way valves are flat faced ANSI flanges, 4-Way valve are DIN raised flanges and require mating DIN weld raised flanges and gaskets. See page 2 for ordering information.

**4-Way Valve**

| Code No.        | Reference | Connection         | Cv* |
|-----------------|-----------|--------------------|-----|
| <b>065B6150</b> | 4 F 50    | 2" DIN Flanged     | 75  |
| <b>065B6165</b> | 4 F 65    | 2-1/2" DIN Flanged | 105 |
| <b>065B6180</b> | 4 F 80    | 3" DIN Flanged     | 175 |
| <b>065B6200</b> | 4 F 100   | 4" DIN Flanged     | 265 |
| <b>065B6225</b> | 4 F 125   | 5" DIN Flanged     | 325 |
| <b>065B6250</b> | 4 F 150   | 6" DIN Flanged     | 465 |

\*Cv = Flow rate allowed by the valve in GPM with a pressure drop of one psi.

**Ordering  
Information (Cont.):**
**DIN Flanges and Gaskets**

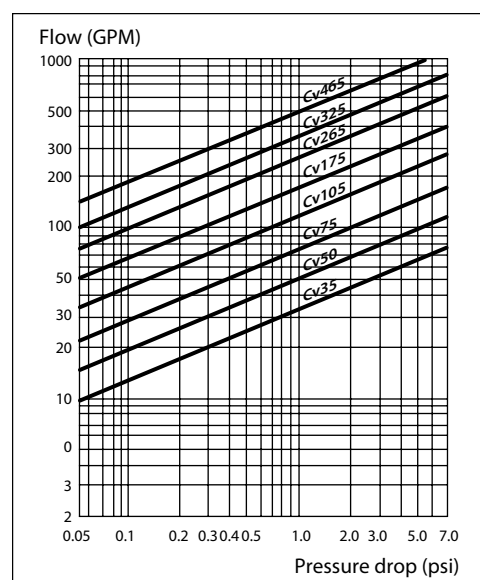
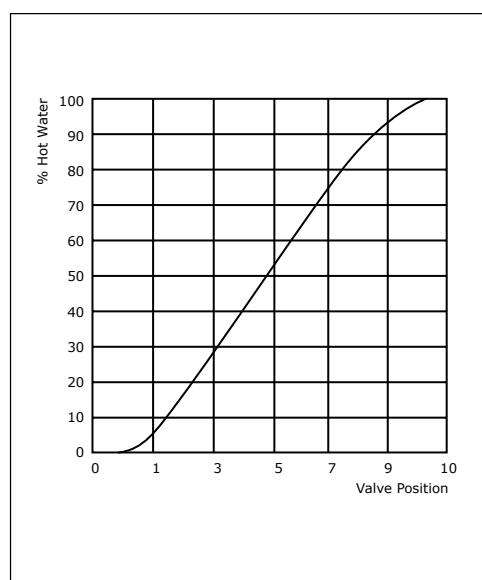
| Code No. | Size   | Description                                                              |
|----------|--------|--------------------------------------------------------------------------|
| 065F8950 | 2"     | DIN Weld Mating Raised Flange<br>(Four Flanges Required per 4-Way Valve) |
| 082F8961 | 2-1/2" |                                                                          |
| 082F8962 | 3"     |                                                                          |
| 082F8963 | 4"     |                                                                          |
| 082F8964 | 5"     |                                                                          |
| 082F8965 | 6"     |                                                                          |
| 065F8951 | 2"     | Gasket for DIN Flange<br>(Four Flanges Required per 4-Way Valve)         |
| 082F8966 | 2-1/2" |                                                                          |
| 082F8967 | 3"     |                                                                          |
| 082F8968 | 4"     |                                                                          |
| 082F8969 | 5"     |                                                                          |
| 082F8970 | 6"     |                                                                          |

**Replacement Parts**

| Code No. | Description       | Valve Sizes | Valve Type |
|----------|-------------------|-------------|------------|
| 065B0003 | Gasket Set        | 2" – 2-1/2" | 3 & 4-Way  |
| 065B0004 |                   | 3" – 6"     |            |
| 082F8981 | Complete Interior | 2-1/2"      | 3-Way      |
| 082F8982 |                   | 3"          |            |
| 082F8983 |                   | 4"          |            |
| 082F8984 |                   | 5"          |            |
| 082F8985 |                   | 6"          |            |
| 082F8987 | Complete Interior | 2"          | 4-Way      |
| 082F8988 |                   | 2-1/2"      |            |
| 082F8989 |                   | 3"          |            |
| 082F8990 |                   | 4" - 5"     |            |
| 082F8991 |                   | 6"          |            |

## Technical Specifications:

|                                                                               |                                                                                                                                                                        |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Max. operating pressure</b>                                                | 90 psi (6 bar)                                                                                                                                                         |
| <b>Max. operating temperature</b>                                             | 230°F (110°C)                                                                                                                                                          |
| <b>Min. operating temperature</b>                                             | 14°F (-10°C)                                                                                                                                                           |
| <b>Max. differential pressure</b>                                             | 1-1/2" – 2": 8 psi (50kPa)<br>2-1/2" – 6": 5 psi (30kPa)                                                                                                               |
| <b>Required motor torque<br/>(Recommendation for<br/>typical application)</b> | Valve size max. 2" – 45 in lb (5Nm)<br>Valve size max. 4" – 85 in lb (10Nm)<br>Valve size max. 6" – 130 in lb (15Nm)<br>(Recommended exercise function for valve body) |
| <b>Allowable fluid</b>                                                        | Water (maximum 50% glycol) in a closed loop hydronic system.                                                                                                           |
| <b>Connections</b>                                                            | 3-Way ANSI Flange, flat faced<br>4-Way DIN Flange, raise faced                                                                                                         |
| <b>Max. Leakage in % of Flow</b>                                              | 1.5                                                                                                                                                                    |



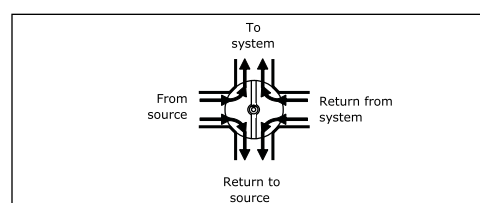
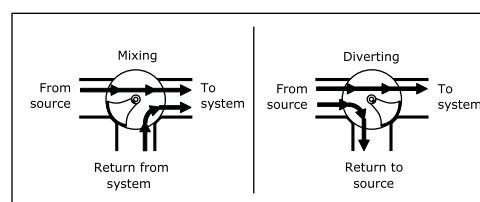
## Piping Arrangement:

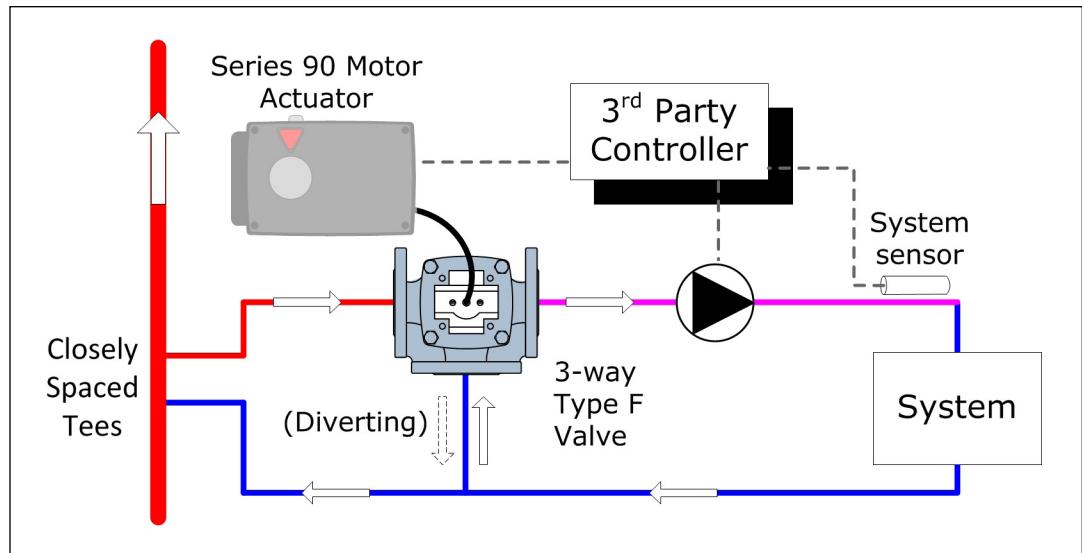
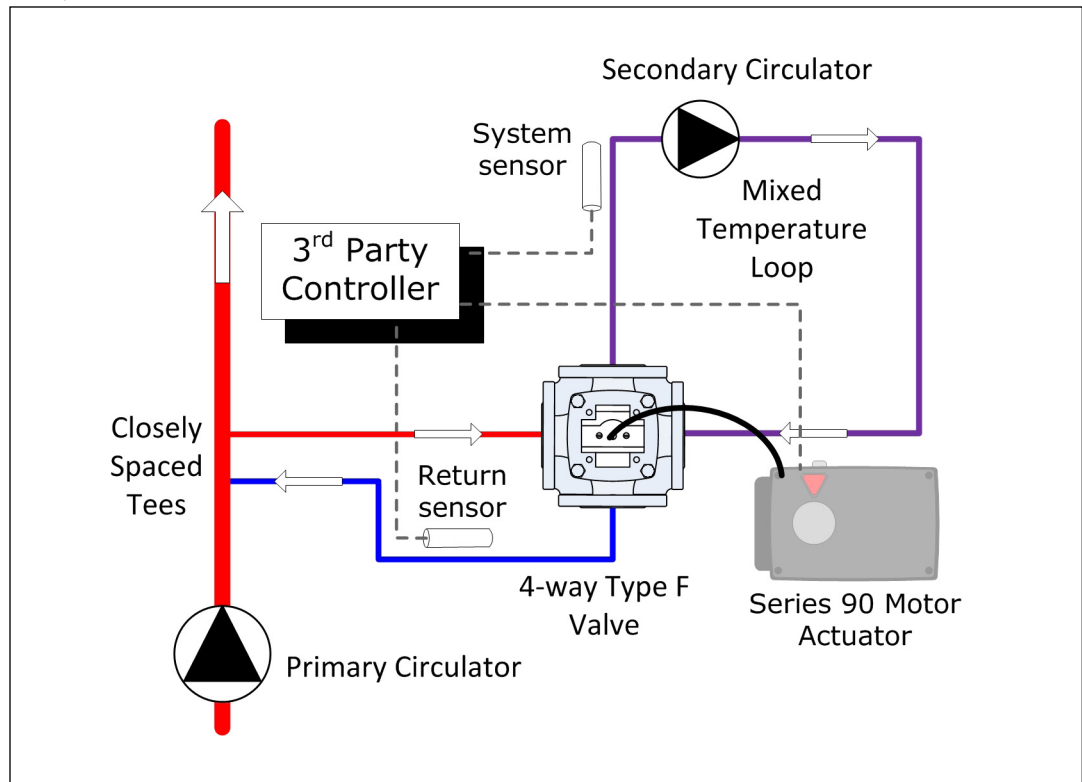
### 3-Way Valves

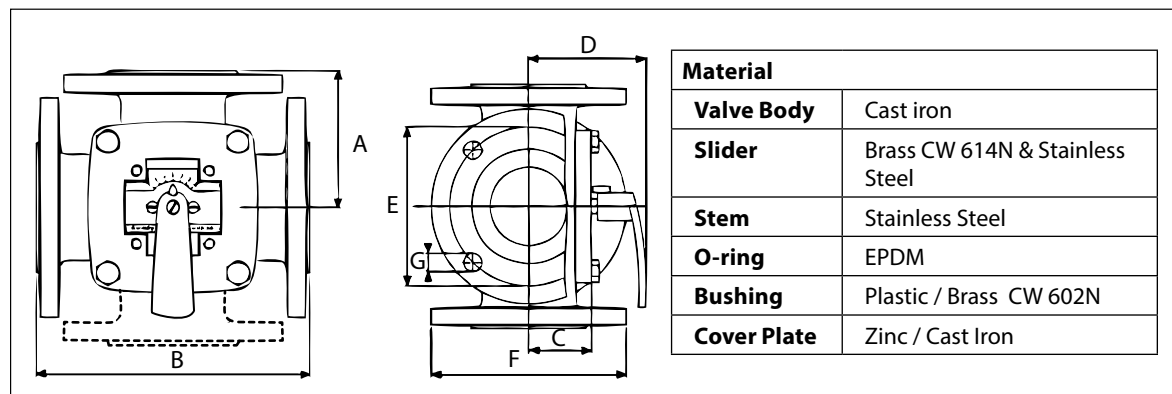
- Ideal valves for mixing and diverting applications
- Reduces risk of low temperature water returning to boiler

### 4-Way Valves

- Double mixing function, providing proportional mixing from source to system and system to source
- Reduces risk of low temperature water returning to boiler.



**Typical Installation****3-Way Valves****4-Way Valves**

**Dimensions:**

**Type F 3-Way, in (mm)**

| Size | Reference | A           | B          | C        | D         | E           | F          | G        | Weight<br>lb (kg) |
|------|-----------|-------------|------------|----------|-----------|-------------|------------|----------|-------------------|
| 2-½" | 3 F65     | 3.9 (100)   | 7.9 (200)  | 2.0 (52) | 3.7 (95)  | 5.5 (139.7) | 7.1 (180)  | 4 x ¾"   | 30.4 (13.8)       |
| 2-½" | 3 F65-50  | 3.9 (100)   | 7.9 (200)  | 2.0 (52) | 3.7 (95)  | 5.5 (139.7) | 7.1 (180)  | 4 x ¾"   | 30.4 (13.8)       |
| 3"   | 3 F80     | 4.6 (117.5) | 9.3 (235)  | 2.5 (63) | 4.2 (106) | 6.0 (152.4) | 7.5 (190)  | 4 x ¾"   | 39.2 (17.8)       |
| 4"   | 3 F100    | 5.2 (132.5) | 10.4 (265) | 2.9 (73) | 4.6 (116) | 7.5 (190.6) | 9.1 (230)  | 8 x ¾"   | 62.4 (28.3)       |
| 5"   | 3 F125    | 5.9 (150)   | 11.8 (300) | 3.1 (80) | 4.8 (123) | 8.5 (215.9) | 10.0 (255) | 8 x 7/8" | 75.2 (34.1)       |
| 6"   | 3 F150    | 6.9 (175)   | 13.8 (350) | 3.5 (88) | 5.1 (130) | 9.5 (241.3) | 11.0 (280) | 8 x 7/8" | 97.2 (44.1)       |

**Type F 4-Way, in (mm)**

| Size | Reference | A         | B          | C        | D         | E         | F          | G        | Weight<br>lb (kg) |
|------|-----------|-----------|------------|----------|-----------|-----------|------------|----------|-------------------|
| 2"   | 4 F50     | 3.9 (98)  | 7.7 (195)  | 2.0 (50) | 3.6 (92)  | 4.3 (110) | 5.5 (140)  | 4 x 0.6" | 24.3 (11.0)       |
| 2-½" | 4 F65     | 3.9 (100) | 7.9 (200)  | 2.0 (50) | 3.6 (92)  | 5.1 (130) | 6.3 (160)  | 4 x 0.6" | 26.9 (12.2)       |
| 3"   | 4 F80     | 4.7 (120) | 9.4 (240)  | 2.6 (65) | 4.3 (108) | 5.9 (150) | 7.5 (190)  | 4 x 0.7" | 44.1 (20.0)       |
| 4"   | 4 F100    | 5.2 (132) | 10.4 (265) | 3.2 (81) | 4.9 (124) | 6.7 (170) | 8.3 (210)  | 4 x 0.7" | 55.1 (25.0)       |
| 5"   | 4 F125    | 5.9 (150) | 11.8 (300) | 3.2 (81) | 4.9 (124) | 7.9 (200) | 9.4 (240)  | 8 x 0.7" | 77.2 (35.0)       |
| 6"   | 4 F150    | 6.9 (175) | 13.8 (350) | 3.5 (89) | 5.2 (131) | 8.9 (225) | 10.4 (265) | 8 x 0.7" | 99.2 (45.0)       |



**Technical Specifications:**

The 3 or 4-Way rotary valve shall be made of a cast iron material with flanged connections. The valve should be capable of being utilized in heating or cooling closed loop applications. The valve position shall have a face plate with printing on both sides to provide a choice of

valve mounting positions. The valve should be capable of assembly with an ESBE series 60 or 90 electric motor actuator.

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# ESBE Series VRG 3 & 4-Way Mixing & Diverting Valves



## Applications:

The ESBE VRG series of 3 & 4-way valves are designed to mix water to the appropriate temperature to meet the zone or system requirements within heating and cooling systems.

The redesigned VRG valve replaces the MG and G series of rotary valves providing additional operational features. By either mixing or diverting, the fluid temperature leaving the valve is regulated by the VRG in either a manual operation or automatically when coupled with the ARA600 Series or 90 Series electronic motor actuator. These actuators are typically connected to an outdoor reset controller, DDC system or boilers with integrated controls.

## Features:

- VRG series valves are a rotary design allowing for great flexibility in field piping layout
- Compact valve size
- Exceptionally low internal leakage rate
- Visible position indicator and knob
- Easily adaptable with ESBE electronic motor actuator styles ARA600 or series 90
- Suitable fluid: water (maximum 50% glycol) for closed loop heating or chilled systems

## Ordering Information:

### 3-Way Valve, Series 131

| Code No. | Size | Connection | Cv <sup>1</sup> | Replaces |            |
|----------|------|------------|-----------------|----------|------------|
|          |      |            |                 | Code No. | Series     |
| 193B1500 | ½"   | Female NPT | 2.9             | 065B8970 | 3MG 15-2.5 |
| 193B1501 | ¾"   |            | 4.7             | 065B8971 | 3MG 20-4   |
| 193B1502 | ¾"   |            | 7.3             | 065B8950 | 3MG 20-6.3 |
| 193B1503 | 1"   |            | 7.3             | 065B8972 | 3MG 25-8   |
| 193B1504 | 1"   |            | 11.7            | 065B8968 | 3MG 25-12  |
| 193B1505 | 1-¼" |            | 18.7            | 065B8969 | 3MG 32-18  |
| 193B1506 | 1-½" |            | 29.3            | 065B8953 | 3 G 40     |
| 193B1507 | 2"   |            | 46.8            | 065B8954 | 3 G 50     |

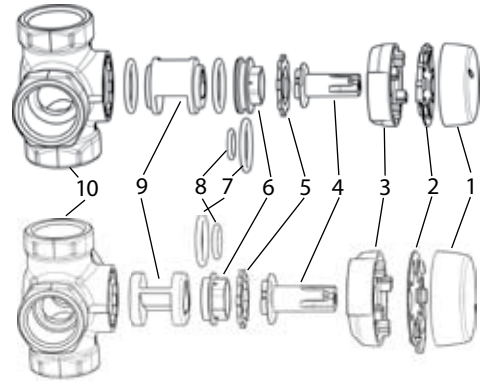
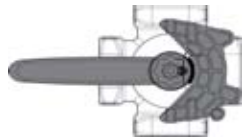
### 4-Way Valve, Series 141

| Code No. | Size | Connection    | Cv <sup>1</sup> | Replaces |            |
|----------|------|---------------|-----------------|----------|------------|
|          |      |               |                 | Code No. | Series     |
| 193B1531 | ¾"   | Female NPT    | 4.7             | 065B8982 | 4MG 20-4   |
| 193B1532 | ¾"   |               | 7.3             | 065B8983 | 4MG 20-6.3 |
| 193B1533 | 1"   |               | 11.7            | 065B8984 | 4MG 25-8   |
| 193B1534 | 1-¼" |               | 18.7            | 065B8991 | 4MG 32-12  |
| 193B1535 | 1-½" |               | 29.3            | 065B8979 | 4 G 40     |
| 193B1536 | 2"   |               | 46.8            | 065B8980 | 4 G 50     |
| 193B1537 | ¾"   | Female Solder | 2.9             | 065B8967 | 4MG 20-2.5 |
| 193B1538 | 1"   |               | 7.3             | 065B8990 | 4MG 25-6.3 |
| 193B1539 | 1-¼" |               | 13.9            | 065B8991 | 4MG 32-12  |

<sup>1</sup>Cv value based on flow per 1 psi pressure drop through the valve.

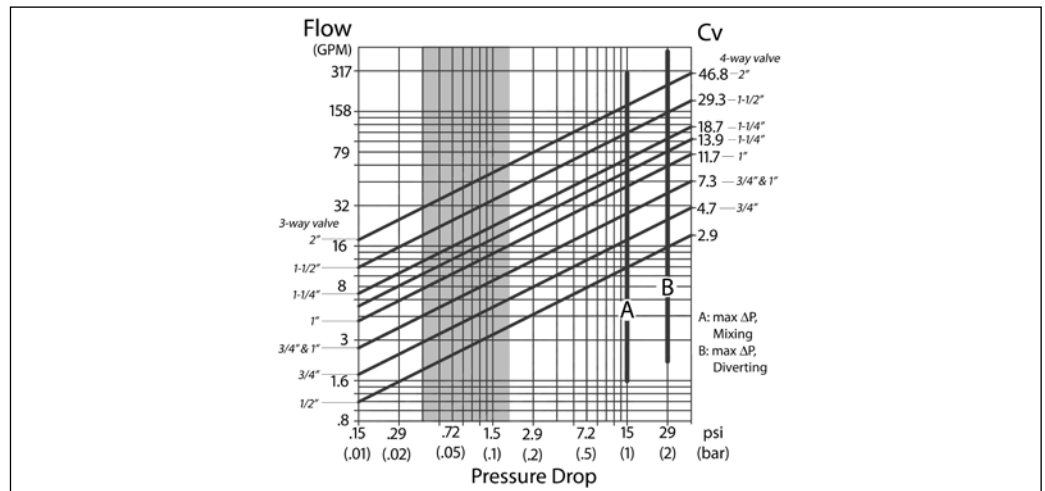
1-1/2" required for Area 7

**Ordering Information**  
**(Cont.):**
**Parts & Accessories**

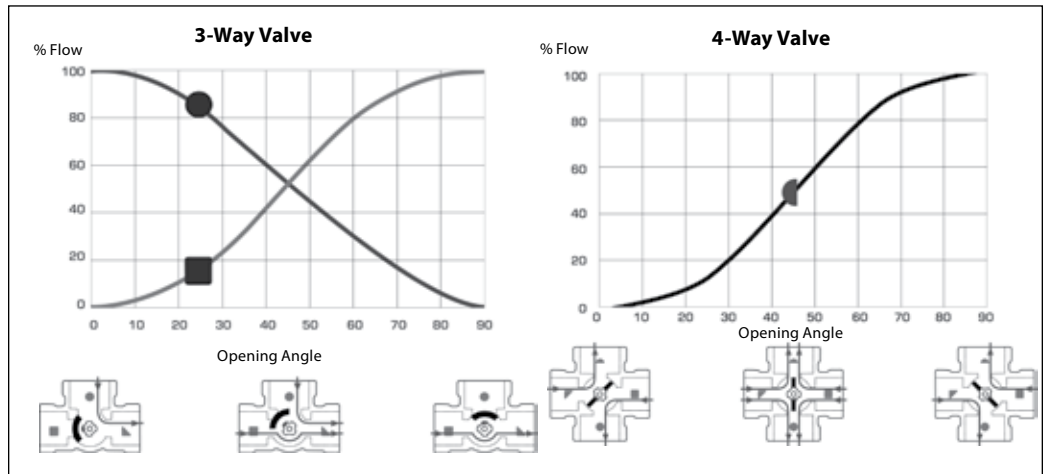
|   | Code No. | Part #'s    | Description                                                                                                    |
|------------------------------------------------------------------------------------|----------|-------------|----------------------------------------------------------------------------------------------------------------|
|                                                                                    | 17006940 | 1 & 2       | VRG valve knob & scale                                                                                         |
|                                                                                    | 17006900 | 3, 4, 6 & 8 | Valve shaft & seal kit,<br>1/2" & 3/4" 3 & 4-way                                                               |
|                                                                                    | 17006910 | 3, 4, 6 & 8 | Valve shaft & seal kit,<br>1" 3 & 4-way                                                                        |
|                                                                                    | 17006920 | 3, 4, 6 & 8 | Valve shaft & seal kit,<br>1-1/4" 3 & 4-way                                                                    |
|                                                                                    | 17006930 | 3, 4, 6 & 8 | Valve shaft & seal kit,<br>1-1/2" & 2" 3 & 4-way                                                               |
|   | 193B1616 | -           | VRG mounting kit for<br>Series 90 actuator                                                                     |
|  | 16000500 | -           | ARA600 motor series<br>mounting kit<br>(contains drive sleeve,<br>pin, rotation indicator &<br>mounting screw) |

**Technical Specifications:**

|                                                                             |                                                    |                |
|-----------------------------------------------------------------------------|----------------------------------------------------|----------------|
| <b>Max. operating pressure</b>                                              | 145psi (10 bar)                                    |                |
| <b>Max. operating temperature</b>                                           | 230°F (110°C)                                      |                |
| <b>Min. operating temperature</b>                                           | -15°F (-10°C)                                      |                |
| <b>Max. differential pressure</b>                                           | 3-way                                              | 4-way          |
|                                                                             | Mixing: 14.5 psi (1bar)<br>Diverting: 20psi (2bar) | 14.5psi (1bar) |
| <b>Leakage in % of flow</b><br>(based on diff. pressure of 14.5psi (1 bar)) | Mixing: <0.05%<br>Diverting: <0.02%                | <1.0%          |
| <b>Max. torque</b>                                                          | <44lbf·in (<5Nm)                                   |                |
| <b>Allowable fluid</b>                                                      | 1) Water, 2) Glycol mixture, max. 50%              |                |

**Pressure Drop Graph:**


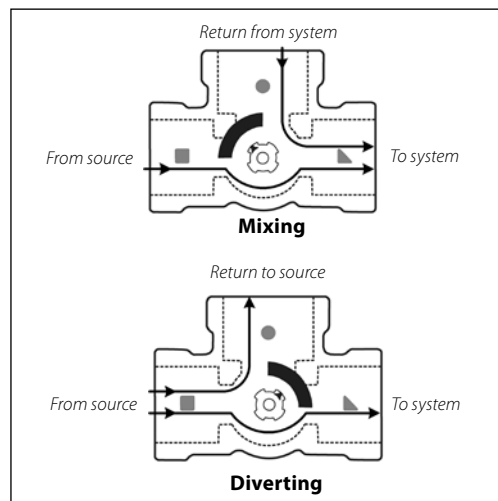
**Flow Characteristics:**



**Piping Arrangement:**

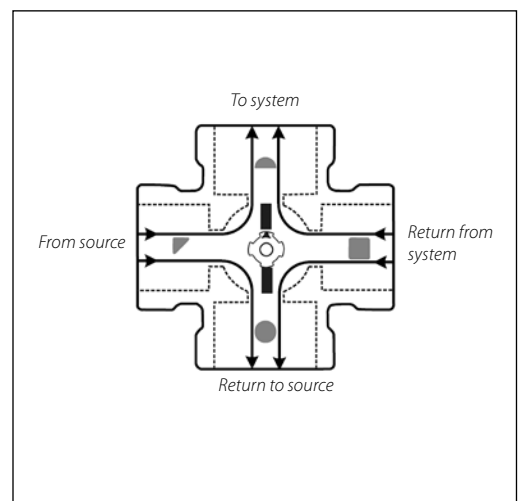
**3-Way valves**

- Ideal valves for mixing and diverting applications
- Reduces risk of low temperature water returning to boiler



**4-Way valves**

- Double mixing function, providing proportional mixing from source to system to source
- Reduces risk of low temperature water returning to boiler

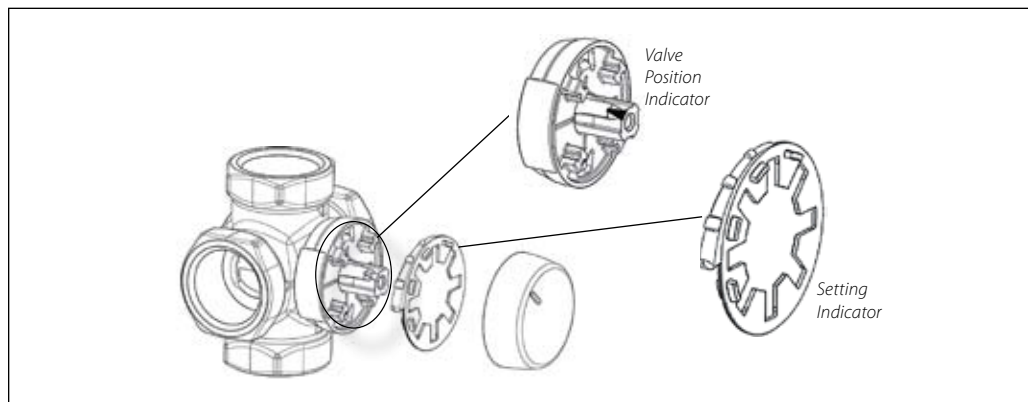
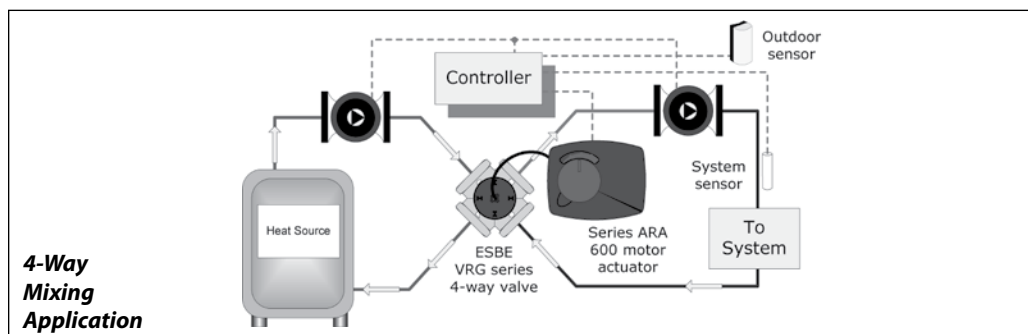
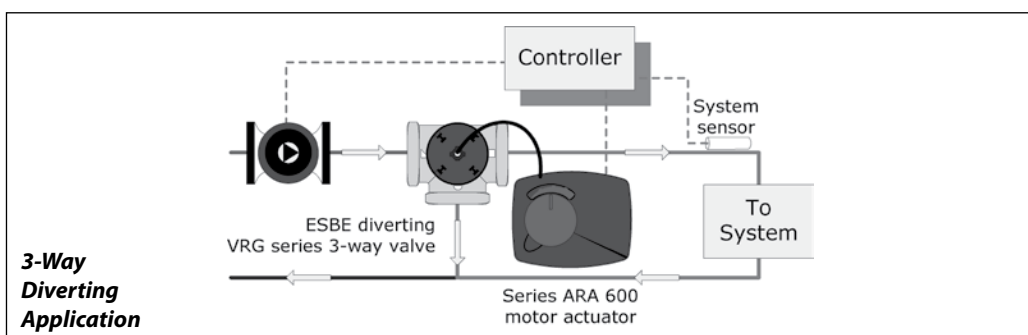
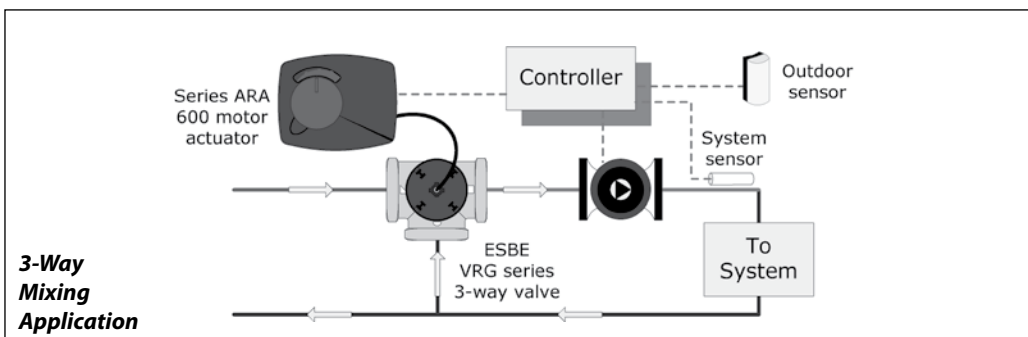


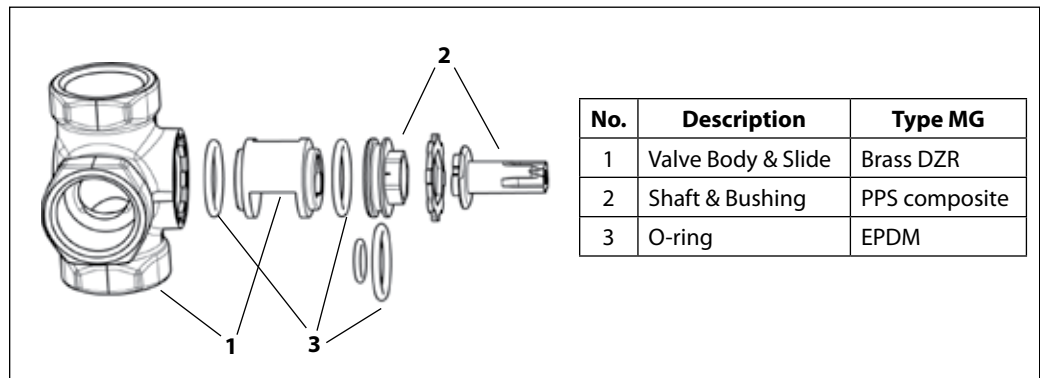
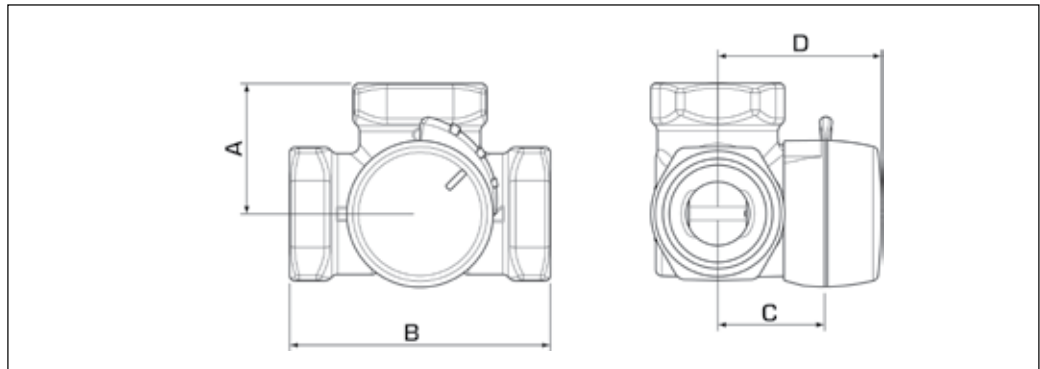
Each port of the VRG valve is stamped with an identifiable shape. These distinct stamped shapes provide a reference during the installation and trouble shooting of the valve.

**Valve Position:**

Located on the stem of the VRG is a position indicator in the shape of an arrow. This arrow correlates to the orientation of the internals of the valve. Depending on the required 90° travel of

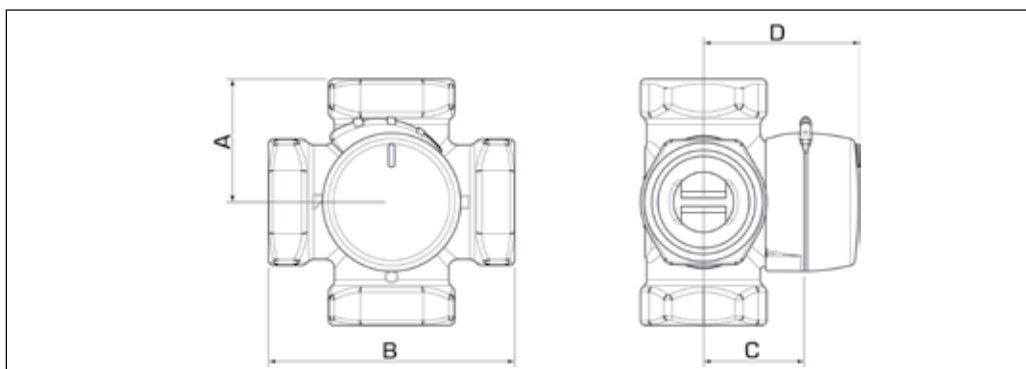
the valve from port to port, the setting indicator can be installed to reflect the direction of travel.

**Typical Installation:**

**Construction:****Dimensions:****VRG 3-Way Valve, Series 131**

| Code Number     | Connection |      | A    |    | B    |     | C    |    | D    |    | Weight |      |
|-----------------|------------|------|------|----|------|-----|------|----|------|----|--------|------|
|                 | Size       | Type | inch | mm | inch | mm  | inch | mm | inch | mm | lb.    | kg   |
| <b>193B1500</b> | 1/2"       | NPT  | 1.42 | 36 | 2.83 | 72  | 1.26 | 32 | 1.97 | 50 | 0.88   | 0.40 |
| <b>193B1501</b> | 3/4"       |      | 1.42 | 36 | 2.83 | 72  | 1.26 | 32 | 1.97 | 50 | 0.95   | 0.43 |
| <b>193B1502</b> | 3/4"       |      | 1.42 | 36 | 2.83 | 72  | 1.26 | 32 | 1.97 | 50 | 0.95   | 0.43 |
| <b>193B1503</b> | 1"         |      | 1.61 | 41 | 3.23 | 82  | 1.34 | 34 | 2.05 | 52 | 1.54   | 0.70 |
| <b>193B1504</b> | 1"         |      | 1.61 | 41 | 3.23 | 82  | 1.34 | 34 | 2.05 | 52 | 1.54   | 0.70 |
| <b>193B1505</b> | 1-1/4"     |      | 1.85 | 47 | 3.70 | 94  | 1.46 | 37 | 2.17 | 55 | 2.09   | 0.95 |
| <b>193B1506</b> | 1-1/2"     |      | 2.09 | 53 | 4.17 | 106 | 1.73 | 44 | 2.36 | 60 | 3.70   | 1.68 |
| <b>193B1507</b> | 2"         |      | 2.36 | 60 | 4.72 | 120 | 1.81 | 46 | 2.52 | 64 | 5.07   | 2.30 |

## Dimensions (Cont.):



VRG 4-Way Valve, Series 141

| Code Number     | Connection |        | A    |    | B    |     | C    |    | D    |    | Weight |      |
|-----------------|------------|--------|------|----|------|-----|------|----|------|----|--------|------|
|                 | Size       | Type   | inch | mm | inch | mm  | inch | mm | inch | mm | lb.    | kg   |
| <b>193B1531</b> | ¾"         | NPT    | 1.42 | 36 | 2.83 | 72  | 1.26 | 32 | 1.97 | 50 | 1.15   | 0.52 |
| <b>193B1532</b> | ¾"         |        | 1.42 | 36 | 2.83 | 72  | 1.26 | 32 | 1.97 | 50 | 1.15   | 0.52 |
| <b>193B1533</b> | 1"         |        | 1.61 | 41 | 3.23 | 82  | 1.34 | 34 | 2.05 | 52 | 1.76   | 0.80 |
| <b>193B1534</b> | 1-¼"       |        | 1.85 | 47 | 3.70 | 94  | 1.46 | 37 | 2.17 | 55 | 2.38   | 1.08 |
| <b>193B1535</b> | 1-½"       |        | 2.09 | 53 | 4.17 | 106 | 1.73 | 44 | 2.36 | 60 | 4.17   | 1.89 |
| <b>193B1536</b> | 2"         |        | 2.36 | 60 | 4.72 | 120 | 1.81 | 46 | 2.52 | 64 | 5.62   | 2.55 |
| <b>193B1537</b> | ¾"         | Solder | 2.44 | 62 | 4.88 | 124 | 1.26 | 32 | 1.97 | 50 | 1.15   | 0.52 |
| <b>193B1538</b> | 1"         |        | 2.56 | 65 | 5.12 | 130 | 1.34 | 34 | 2.05 | 52 | 1.76   | 0.80 |
| <b>193B1539</b> | 1-¼"       |        | 3.03 | 77 | 6.06 | 154 | 1.46 | 37 | 2.17 | 55 | 2.38   | 1.08 |

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**Features:**



The ESBE 24 volt electronic series of motors are direct mounted actuators that rotate ESBE 3 & 4-way rotary valves for hydronic heating and cooling systems. The Series 90 actuator provides additional torque compared to the Series 60 and ARA series of actuators, and can be installed on all ESBE 3 & 4 way rotary valves, in sizes from 1/2" – 6" valves. The Series 90 motors also have linkage kits available capable of adapting to Viessmann, Honeywell, Centra DR/ZR, Landis & Staefa and TA VTR styles of valves.

**Features:**

- ESBE control motor series are of a compact design
- Direct mount actuators
- Easy manual override of motor actuator
- Adjustable rotation between 30° and 180° with internal cams
- Floating 3-point and proportional signal operation
- Several motors with an auxiliary contact switch activated through an adjustable cam

**Ordering information:**

**90 Series Actuator: All ESBE rotary mixing valves** (91EM, valves up to 2" only for diverting application)

065F8952 Required

| Code No. | Series | Signal                                | Run Time<br>(90°@60Hz) |
|----------|--------|---------------------------------------|------------------------|
| 065F8952 | 91EM   | Floating 3 point,<br>with Aux. Switch | 12 sec.                |
| 065F8953 | 92EM   |                                       | 50 sec.                |
| 065F8959 | 92-2EM |                                       | 100 sec.               |
| 065F8954 | 93EM   |                                       | 4 min.                 |
| 065F8955 | 92P    | Proportional                          | 60, 90 or 120 sec.     |

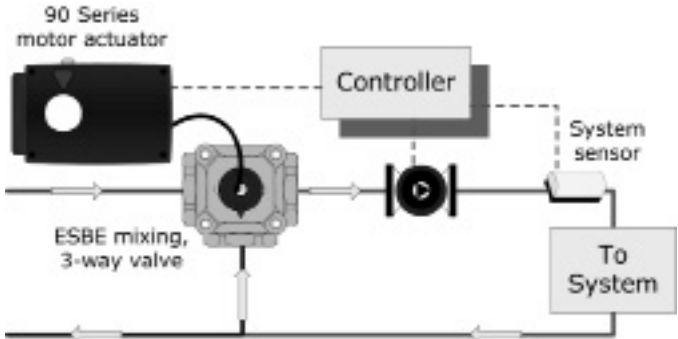
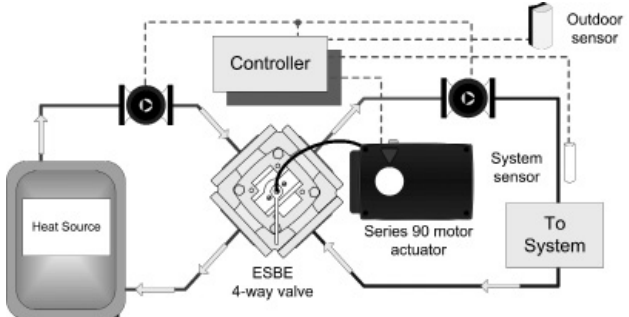
\*If the series 90 actuator will be mounted to a VRG series of valve body an adapter mounting kit is required (Code No. 193B1616)

**Replacement Parts:**

**90 Series Actuator**

|  | Code No. | Description                            | Item #      |
|--|----------|----------------------------------------|-------------|
|  | 25413000 | Metal drive sleeve                     | 2           |
|  | 193B1615 | Mounting kit for 90 series             | 1-10        |
|  | 193B1616 | Series 90 actuators to ESBE VRG valves | (Not shown) |

**Applications:**

| 3-way mixing valve application                                                     |                                                                                                                                                              |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p>When coupled with a controller the assembly of the 90 Series motor actuator and ESBE rotary valve provide regulation of the fluid through the system.</p> |
| 4-way mixing valve application                                                     |                                                                                                                                                              |
|  | <p>The Series 90 motor actuator provides higher torque capabilities allowing use on larger sized valves requiring up to 135 in. lbs. of torque.</p>          |

**Accessories:**

| Linkage Kit     | Non ESBE Valves       |
|-----------------|-----------------------|
| <b>065F8957</b> | Viessmann Valve       |
| <b>065F8958</b> | Centra #900 Valve     |
| <b>065F8990</b> | Landis & Staefa Valve |
| <b>065F8991</b> | Oventrop Valve        |
| <b>065F8992</b> | TA VTR Valve          |

**3 & 4-way Mixing Valves:**
**3-way Valves**

| Code No.        | Reference | Connection   | Cv <sup>0</sup> |
|-----------------|-----------|--------------|-----------------|
| <b>193B1500</b> | VRG 131   | ½", FNPT     | 2.9             |
| <b>193B1501</b> | VRG 131   | ¾", FNPT     | 4.7             |
| <b>193B1502</b> | VRG 131   | ¾", FNPT     | 7.3             |
| <b>193B1503</b> | VRG 131   | 1", FNPT     | 7.3             |
| <b>193B1504</b> | VRG 131   | 1", FNPT     | 11.7            |
| <b>193B1505</b> | VRG 131   | 1-¼", FNPT   | 18.7            |
| <b>193B1506</b> | VRG 131   | 1-½", FNPT   | 29.3            |
| <b>193B1507</b> | VRG 131   | 2", FNPT     | 46.8            |
| <b>065B8960</b> | 3 F 65-50 | 2-½" Flanged | 75              |
| <b>065B8961</b> | 3 F 65    | 2-½" Flanged | 105             |
| <b>065B8962</b> | 3 F 80    | 3" Flanged   | 175             |
| <b>065B8963</b> | 3 F 100   | 4" Flanged   | 265             |
| <b>065B8964</b> | 3 F 125   | 5" Flanged   | 325             |
| <b>065B8965</b> | 3 F 150   | 6" Flanged   | 465             |

<sup>0</sup>Cv value in GPM at pressure drop of 1 psi

Contact Danfoss for information on MG or G styles of valves

**4-way Valves**

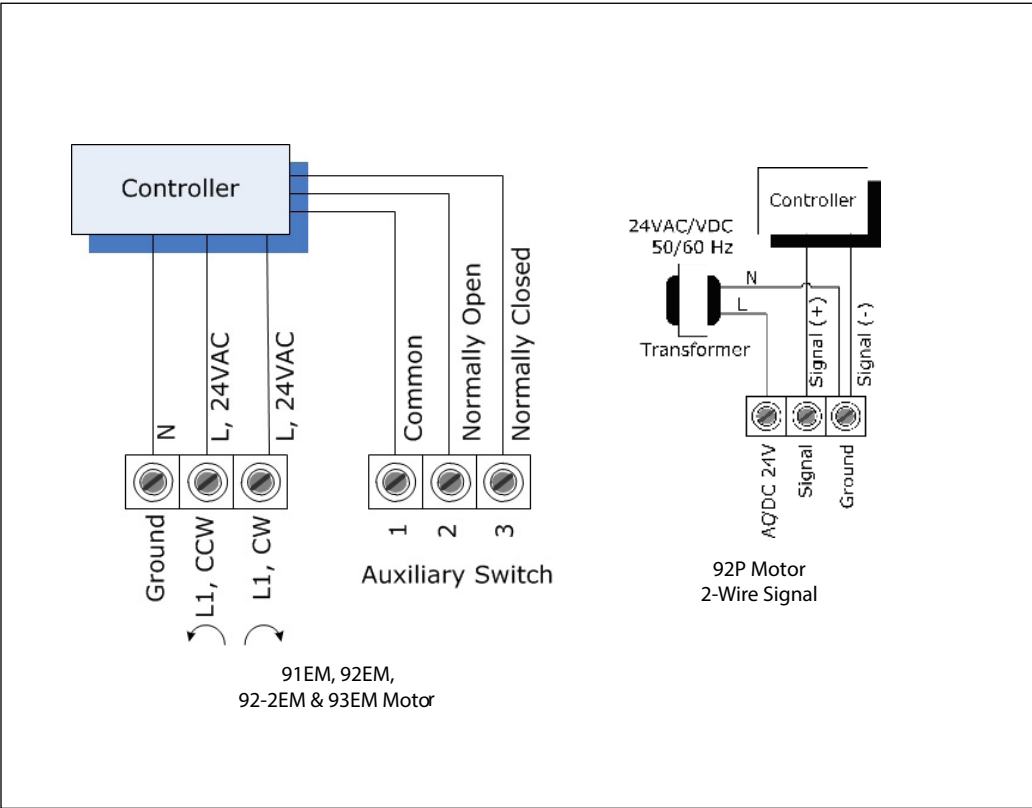
| Code No.        | Reference | Connection   | Cv <sup>0</sup> |
|-----------------|-----------|--------------|-----------------|
| <b>193B1531</b> | VRG 141   | ¾", FNPT     | 4.7             |
| <b>193B1532</b> | VRG 141   | ¾", FNPT     | 7.3             |
| <b>193B1533</b> | VRG 141   | 1", FNPT     | 11.7            |
| <b>193B1534</b> | VRG 141   | 1-¼", FNPT   | 18.7            |
| <b>193B1535</b> | VRG 141   | 1-½", FNPT   | 29.3            |
| <b>193B1536</b> | VRG 141   | 2", FNPT     | 46.8            |
| <b>193B1537</b> | VRG 141   | ¾", Solder   | 2.9             |
| <b>193B1538</b> | VRG 141   | 1", Solder   | 7.3             |
| <b>193B1539</b> | VRG 141   | 1-¼", Solder | 13.9            |
| <b>065B6150</b> | 4 F 50    | 2" Flanged   | 75              |
| <b>065B6165</b> | 4 F 65    | 2-½" Flanged | 105             |
| <b>065B6180</b> | 4 F 80    | 3" Flanged   | 175             |
| <b>065B6200</b> | 4 F 100   | 4" Flanged   | 265             |
| <b>065B6225</b> | 4 F 125   | 5" Flanged   | 325             |
| <b>065B6250</b> | 4 F 150   | 6" Flanged   | 465             |

**Technical Specifications:**

|                                               |                                                                                             |         |                                                                                                                                      |
|-----------------------------------------------|---------------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Supply Voltage</b>                         | 24VAC 50/60 Hz,<br>(92P: 24VAC/DC 50/60 Hz)                                                 |         |                                                                                                                                      |
| <b>Power Consumption</b>                      | 91EM / 92EM / 92-2EM / 93EM: 2VA                                                            |         |                                                                                                                                      |
| <b>Rotation (Factory set)</b>                 | 90°, adjustable b/w 30° and 180° (92P only 90° rotation)                                    |         |                                                                                                                                      |
| <b>Torque</b>                                 | 135 in. lb (15Nm)<br>[91EM: 45 in. lb (5Nm)]                                                |         |                                                                                                                                      |
| <b>Mountable Valve Styles and Valve Sizes</b> | ESBE rotary mixing valves MG / G / F / VRG styles<br>Valve Sizes ½" to 6", (91EM: Up to 2") |         |                                                                                                                                      |
| <b>Enclosure</b>                              | NEMA 13, IP 54                                                                              |         |                                                                                                                                      |
| <b>Ambient Temp. Range</b>                    | 5 to 131°F (-15 to 55°C)                                                                    |         |                                                                                                                                      |
| <b>Weight</b>                                 | 1.8lb (0.8kg)                                                                               |         |                                                                                                                                      |
| <b>Signal / Run Time</b>                      | <b>3-point floating:</b>                                                                    |         | <b>Proportional:</b> (selectable)<br>92P-<br>Signal:<br>0-10V, 2-10V, 1-5V,<br>0-20mA, 4-20mA<br><br>Run Time:<br>60, 90 or 120 sec. |
|                                               |                                                                                             | 50 Hz   | 60 Hz                                                                                                                                |
|                                               | 91EM                                                                                        | 15 sec. | 12 sec.                                                                                                                              |
|                                               | 92EM                                                                                        | 1 min.  | 50 sec.                                                                                                                              |
|                                               | 92-2EM                                                                                      | 2 min.  | 100sec.                                                                                                                              |
|                                               | 93EM                                                                                        | 5 min.  | 4 min.                                                                                                                               |
| <b>Auxiliary Switch</b>                       | All motors except 92P are with an adjustable cam position, max. 2A                          |         |                                                                                                                                      |



Electrical Wiring:



Installation:

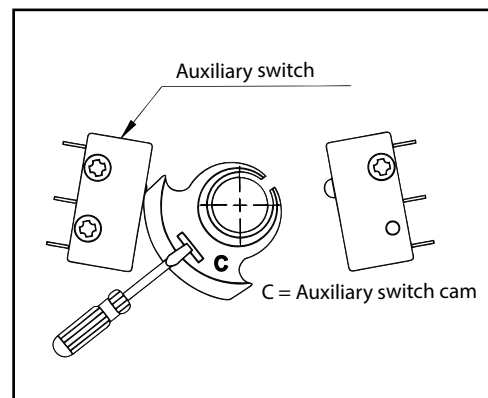
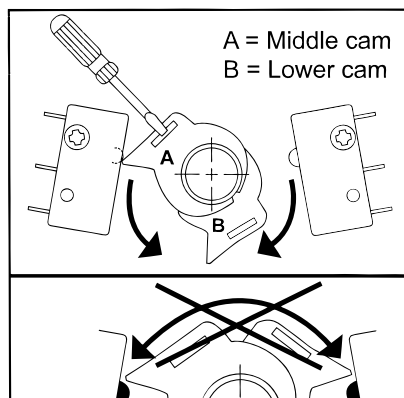
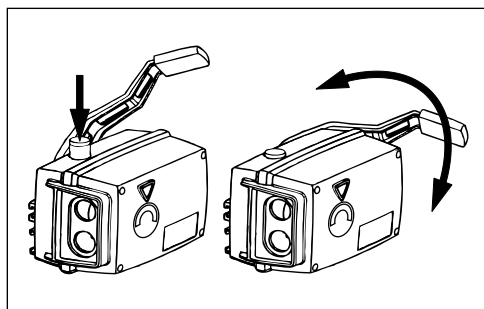
The actuator should be mounted with the valve stem in either a horizontal position or pointing upwards. The actuator should not be installed in a downward orientation.

|                                                                                                                                                                                                                                                                                                                                                           |  |                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                           |  | <ul style="list-style-type: none"><li>1) Threaded stud</li><li>2) Drive sleeve</li><li>3) Locking screw</li><li>4) Handle</li><li>5, 6, 8, 9) Mounting hardware</li><li>7) Mounting base</li></ul> |
| <p>When installing the handle (4), it should be installed opposite to the indicator on the drive sleeve and the end of the handle should be facing away from the valve. Failure to properly install the handle will result in damage to the 90 series motor actuator due to the interference of the mounting stud (1) with the rotation of the handle</p> |  |                                                                                                                                                                                                    |
|                                                                                                                                                                                                                                                                                                                                                           |  |                                                                                                                                                                                                    |

**Adjustment of Rotation  
and Auxiliary Switch Cam:**

Within the 90 Series motors, the factory set 90° of rotation can be adjustable between 30° and 180°. The adjustment of the angle of rotation and the engagement of the auxiliary switch is set

via internal cams within the motor actuator for the 91EM, 92EM, 92-2EM and 93EM. Care should be taken to ensure the cams are engaging the switches properly.


**Manual Operation:**


When placing the actuator in manual operation power should be turned off to the actuator. In the event the actuator is powered and is left in a manual position, the result could lead to the internal damage of the actuator.

When placing the 90 Series motor in manual operation, the gray button must be pushed in. Then the handle can be rotated either clockwise or counter clockwise. To return back to automatic operation, rotate the handle until the button pops out.

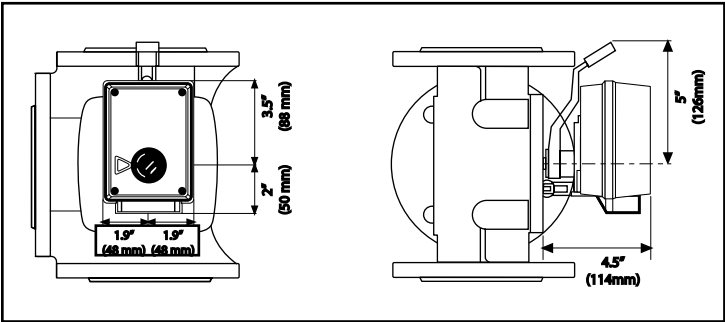


Proportional  
Actuator Settings:

92P Actuator  
Internal settable jumpers

| Actuating time                                                                    |                                                                                   |                                                                                     |                                                                                       |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                                                                                   | 60s                                                                               | 90s                                                                                 | 120s                                                                                  |
| 60s                                                                               | ..                                                                                |    | ..                                                                                    |
| 90s                                                                               | ..                                                                                |    | ..                                                                                    |
| 120s                                                                              | ..                                                                                | ..                                                                                  |    |
| Rotation direction                                                                |                                                                                   |                                                                                     |                                                                                       |
| Clock wise                                                                        |                                                                                   | Counter clock wise                                                                  |                                                                                       |
|  |                                                                                   | ..                                                                                  |                                                                                       |
| ..                                                                                |                                                                                   |  |                                                                                       |
| Control signal                                                                    |                                                                                   |                                                                                     |                                                                                       |
|                                                                                   | 0-10V                                                                             | 2-10V                                                                               | 1-5V                                                                                  |
| OFS                                                                               | ..                                                                                | ..                                                                                  | ..                                                                                    |
| DIV                                                                               |  |    | ..                                                                                    |
| CUR                                                                               | ..                                                                                |    |    |
|                                                                                   |                                                                                   |                                                                                     |                                                                                       |
|                                                                                   | 0-5V                                                                              | 0-20mA                                                                              | 4-20mA                                                                                |
| OFS                                                                               | ..                                                                                | ..                                                                                  |    |
| DIV                                                                               | ..                                                                                | ..                                                                                  | ..                                                                                    |
| CUR                                                                               | ..                                                                                |  |  |

Dimensions:



Typical Specifications:

The electric actuator shall be a compact style, mountable on a rotary valve. The actuator shall be 24Vac powered with an ability for manual override via push in button. It shall have a clear

indication of the actuator's position and be available as either a floating point or proportional style of actuator.

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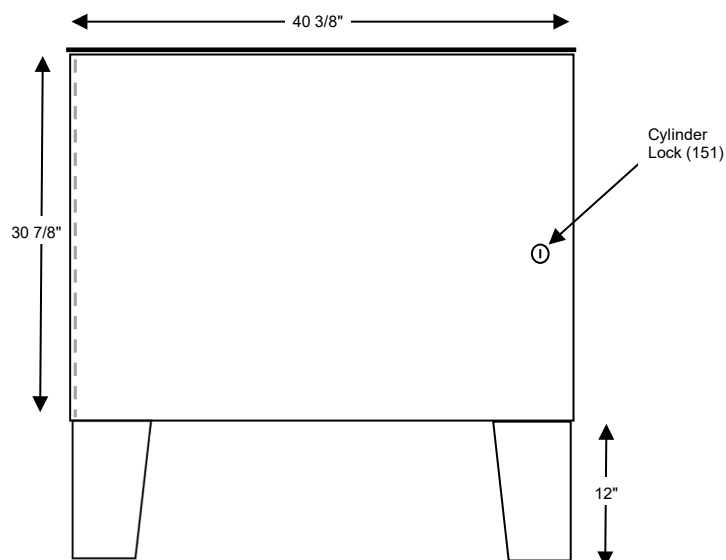
40"w X 30"h X 6"d

## VALVE MANIFOLD CABINET FREE STANDING

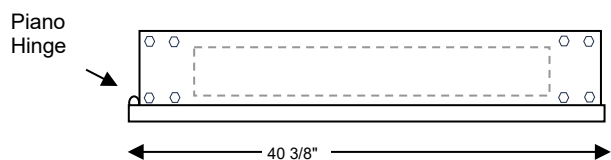
### STANDARD CONSTRUCTION FEATURES:

- Full Metal Door w/ Full Length Stainless Steel Piano Hinge
- 316 Stainless Steel Construction (16 ga.) w/ Satin #4 Finish\*
- Cylinder Lock (keyed alike) w/ Friction Catch (x2)
- 12" Legs (x2) w/ SS Nuts and Bolts
- Open bottom panel (26"w x 4"d)
- Closed Cell Neoprene Weather Gasket
- Rain-Edge Roof
- Manifold mounting holes/plate available upon request

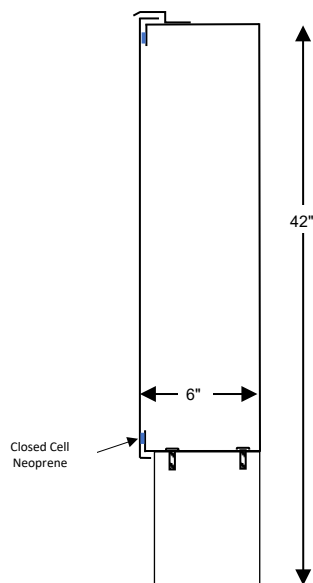
### FRONT



### TOP



### SIDE



\*Drawings not to scale

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